The Algorithmic Liquidity Trap: High-Velocity Swiping vs. Deep Engagement in Short-Video Platforms*

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Abstract

Short-video platforms balance user retention with high-frequency impression generation. We model this ecosystem using a Mean-Field Stackelberg Game with "traffic recycling," where users constantly re-enter the content pool. Unlike much of the theoretical literature that relies solely on stylized parameters, we calibrate the model's micro-foundations using KuaiRand—a large-scale randomized experiment dataset from Kuaishou—ensuring that user behavioral parameters reflect real-world attention dynamics. Solving the coupled HJB and Non-Local Fokker-Planck equations via a novel Spectral Soft Actor-Critic algorithm, we identify a critical "Algorithmic Liquidity Trap." While unconstrained optimization mathematically maximizes swipe flux, it drives dwell times to a physiological floor well below the IAB/MRC viewability standard of 2 seconds. We term this "Phantom Liquidity"—high throughput with zero effective economic value. Furthermore, we reveal a "Dopamine Treadmill" effect: improved content quality paradoxically increases user impatience, explaining the difficulty of sustaining long-form content. Our results demonstrate that without explicit constraints—such as weights for completion rates and creator interactions—algorithmic maximization inevitably collapses the ecosystem, validating the necessity of current industry governance structures.

Keywords: *short-video platforms, mean-field games, traffic recycling, deep reinforcement learning, algorithmic liquidity trap, platform governance.*
JEL Classification: *L86; C73; M37.*

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1 Introduction

In the digital attention economy, short-video platforms such as TikTok, Instagram Reels, and YouTube Shorts have established a dominant paradigm of content consumption: the infinite scroll. Unlike traditional media (e.g., Netflix) that monetizes user retention via subscriptions, or search engines (e.g., Google) that monetize specific intent, short-video platforms operate on a unique hybrid model. They generate revenue through a delicate balance of two conflicting metrics: *duration* (total time spent on the app) and *flux* (the frequency of exposure to new content, primarily ads). This creates a fundamental operational trade-off: To maximize ad inventory, the platform is incentivized to accelerate the "swiping" velocity of users. However, indiscriminate acceleration risks decoupling algorithmic optimization from physiological reality, potentially leading to a state of hollow engagement. The central managerial question, therefore, is: *How should a platform regulate the granular flow of user attention to avoid the "race to the bottom" and balance immediate impression yield against long-term ecosystem sustainability?*

Existing theoretical frameworks in operations management and economics struggle to fully capture the dynamics of this ecosystem due to a structural disconnect between micro-level behavior and macro-level flows. The classical literature on user behavior, rooted in Optimal Stopping Theory (Peskir & Shiryaev, 2006; Stokey, 2008), typically treats the decision-maker as facing a static environment—once a user exercises the option to "stop" (or switch), they exit the system. However, the short-video ecosystem is characterized by a closed-loop mechanism we term *"Traffic Recycling."* When a user swipes away from a video, they do not leave the platform; rather, they are immediately "recycled" back into the content pool to face a new draw. This recycling mechanism implies that the platform's algorithmic interventions do not merely affect a single viewing session but fundamentally reshape the macroscopic equilibrium of the entire user population.

To bridge this gap, this paper develops a *Mean-Field Stackelberg Game (MFSG)* framework that explicitly models the interplay between the platform's dual-objective optimization and the users' continuous-time stopping decisions. We model the platform as a Stackelberg Leader that regulates the ecosystem through two levers: *Retention Control* (micro-interventions to extend the duration of the current video) and *Matching Control* (investment to improve the quality of the next video). Users, acting as rational Followers, observe these macroscopic controls and solve an optimal stopping problem with an endogenous threshold. Mathematically, this leads to a novel coupled system comprising a Hamilton-Jacobi-Bellman (HJB) equation governing user strategies and a *Non-Local* Fokker-Planck equation describing the population evolution with a source term induced by traffic recycling. We solve this high-dimensional control problem using a physics-informed Deep Reinforcement Learning (DRL) algorithm, calibrated with large-scale randomized experiment data (KuaiRand) to ensure empirical realism.

Our analysis yields three distinct insights that challenge the blind pursuit of engagement metrics. First, we identify a *"Dopamine Treadmill"* effect. One might intuitively expect that improving recommendation quality would satisfy users and prolong their stay on individual videos. Contrary to this, we find that a higher quality content pool raises the users' rational expectation of the "next best alternative," thereby increasing their optimal switching threshold. This paradox explains the structural difficulty of sustaining *long-form content* or deep engagement within a pure feed architecture: better matching algorithms inadvertently condition users to be more impatient.

Second, and most critically, we uncover a regime we term the *"Algorithmic Liquidity Trap."* Traditional models often assume that maximizing swipe flux correlates with maximizing value. However, our general equilibrium analysis reveals that under an unconstrained ad-centric strategy, the system converges to a *"High-Velocity Saturation"* point. While mathematically maximizing the traffic recycling loop, this equilibrium drives the average dwell time to a physiological floor far below the IAB/MRC viewability standard of 2 seconds. We define this state as *"Phantom Liquidity"*—a condition of high system throughput but zero effective economic value. This finding serves as a counterfactual proof, validating the necessity of current industry practices that strictly enforce constraints on viewability and dwell time.

Third, we characterize the structure of the optimal algorithmic intervention, which exhibits a non-linear *"Retention Trap."* We find that the optimal policy applies peak intervention intensity only when the user's attention approaches the critical churn threshold. This "Trap and Release" strategy exploits the users' decision boundary to maximize ad extraction. While effective in a static sense, this aggressive extraction creates a "Tragedy of the Commons" risk for the creator ecosystem, further underscoring the need for governance mechanisms—such as algorithmic weighting for *completion rates* and *social interactions*—to align platform incentives with sustainable user value.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 formulates the Mean-Field Stackelberg Game model. Section 4 details the structural calibration using the KuaiRand dataset. Section 5 presents the main equilibrium analysis and the "Algorithmic Liquidity Trap." Section 6 concludes with managerial implications for platform governance.

2 Literature Review

Our work sits at the intersection of three streams of literature: the economics of rational attention (optimal stopping), algorithmic operations in recommender systems, and mean-field game theory.

2.1 Rational Attention and Optimal Stopping

The micro-foundation of our model builds upon the classical theory of optimal stopping under uncertainty. The seminal works of Stokey (2008) and Peskir and Shiryaev (2006) established the mathematical rigor for single-agent stopping problems driven by geometric Brownian motions. In the context of the digital economy, this framework has been adapted to model online search behavior. For instance, Gershkov and Moldovanu (2009) analyze consumer search with learning, while Ke, Shen, and Villas-Boas (2016) propose a Bayesian framework for rapid information consumption.

However, these classical models typically assume an "absorbent" exit state—once a user stops (or purchases), they leave the system. Our work extends this by introducing the concept of *"Traffic Recycling,"* where the "stopping" action (swiping) is not an exit but a transition back into the content pool. This requires a non-local boundary condition that couples the exit flux of one state to the entry flux of another, a dynamic absent in standard search theory.

2.2 Algorithmic Management of Attention in Recommender Systems

The second stream focuses on the operational design of content platforms. In the computer science domain, early works prioritized binary metrics such as Click-Through Rate (CTR). More recently, industrial frameworks like those at YouTube (M. Chen et al., 2019; Covington, Adams, & Sargin, 2016) have shifted towards continuous engagement metrics (e.g., watch time) via Deep Reinforcement Learning.

However, a theoretical gap persists in the Operations Management (OM) and Economics literature regarding the *objective function* of these algorithms. Traditional models in online advertising and revenue management typically posit a *linear relationship* between impression flux and revenue (e.g., Goldfarb & Tucker, 2011; Lobel, Patel, Vulcano, & Zhang, 2016). The implicit assumption is that maximizing the throughput of users (i.e., swipe frequency) monotonically increases ad inventory value, effectively treating attention as a standard divisible good.

Our work challenges this "Linear Flux Hypothesis." We argue that in short-video ecosystems, the value function is a *step function* governed by viewability thresholds (e.g., the IAB 2-second standard). By incorporating this discontinuity, we distinguish between *nominal flux* (server load) and *effective economic flow*, a nuance often overlooked in standard inventory maximization models. This allows us to theoretically characterize the phenomenon of "Phantom Liquidity"—a regime where standard OM policies fail by optimizing for empty calories.

2.3 Mean-Field Games in Platform Economies

Finally, we contribute to the emerging application of Mean-Field Games (MFG) in economics. While traditional heterogeneous-agent frameworks in macroeconomics (Aiyagari, 1995) or cooperative control approaches like potential games (Marden, Arslan, & Shamma, 2009) offer valuable insights for static or discrete-time equilibrium, they often face the "curse of dimensionality" when handling continuous-time attention flows with complex recycling dynamics. The adoption of the MFG framework is particularly natural in the context of short-video platforms due to the sheer scale of the user base. Platforms like TikTok and Instagram host billions of active users, rendering traditional N-player game theoretical models computationally intractable. As the number of agents $N \to \infty$, individual idiosyncratic noises average out, allowing us to approximate the complex high-dimensional system with a tractable interaction between a representative agent and the aggregate population density (Lasry & Lions, 2007).

Since its foundation, MFG has been applied to systemic risk, crowd dynamics, and macroeconomics (Achdou, Han, Lasry, Lions, & Moll, 2022). Recently, Iyer, Johari, and Sundararajan (2014) and Yang et al. (2018) have applied mean-field approximations to large-scale information systems.

Our work introduces a distinct technical novelty to this field: the *Mean-Field Stackelberg Game with Recycling*. Unlike standard MFGs where the population mass is conserved via a simple transport equation or vanishes at the boundary, our model features a "re-entrant" mechanism governed by a Non-Local Fokker-Planck equation. This structure allows us to capture the "velocity" of user flow—a critical metric in short-video platforms that standard MFG models ignore.

3 The Model

We formulate the interaction between the short-video platform and its massive user base as a Mean-Field Stackelberg Game. The platform (Leader) commits to an algorithmic policy to maximize long-term rewards, while individual users (Followers) observe the ecosystem's state and optimally decide when to "swipe" to the next video.

3.1 Empirical Motivation: Stylized Facts

Before formalizing the stochastic dynamics, we summarize three critical stylized patterns observed in the high-frequency interaction data (KuaiRand). These patterns serve as the empirical micro-foundations for our specific design of the attention dynamics and traffic recycling mechanism.

- **Pattern 1: Rapid Initial Filtering (The "Golden Seconds").** The empirical distribution of dwell times reveals a sharp density spike in the interval $[0, 2s]$, indicating that users make rapid, reflex-driven rejection decisions. This motivates our use of an **Absorbing Boundary** at a lower threshold b^* rather than a purely duration-based utility model.

- **Pattern 2: Heavy-Tailed Engagement (The "Gambling" Effect).** Despite the rapid decay, the survival function exhibits a fat tail (high kurtosis). This suggests that attention is driven by high volatility (σ), validating the **Geometric Brownian Motion** assumption where users chase stochastic rewards.

- **Pattern 3: Closed-Loop Circulation (Traffic Recycling).** Analysis of session logs shows that a "swipe" event rarely terminates the session ($P(Exit) \approx 1.25\%$). Instead, it triggers an immediate re-entry. This necessitates a **Non-Local PDE framework** where the outflow mass is reinjected into the system, rather than a standard optimal stopping model with permanent exit.

3.2 The Micro-Ecosystem: Attention Dynamics and Retention Control

Building on these stylized patterns, we now formally define the user's attention process.

Scope of Analysis: The Demand-Side Loop. To isolate the mechanism of the "Algorithmic Liquidity Trap," we focus our analysis on the closed-loop interaction between the *Platform's Algorithmic Policy* and the *User's Attention Dynamics*. We treat the content supply side (creators) as an instrument controllable by the platform via the matching variable $u(t)$, which shifts the quality distribution $g(z; u)$. While we acknowledge that creators inevitably react to algorithmic incentives in the long run (a supply-side response), explicitly modeling the creator game would introduce significant tractability issues without altering the fundamental existence of the liquidity trap driven by user physiology. We thus abstract away strategic creator behavior to focus on the immediate feedback loop between algorithmic acceleration and cognitive capacity.

Assumption (The Infinite Feed Paradigm). We restrict our analysis to the *Feed-based Infinite Scroll* mechanism (e.g., TikTok's "For You" page), characterized by passive consumption and high-frequency switching. We distinguish this from *search-based* or *long-form* consumption (e.g., Netflix), which follows different decision dynamics.

The dynamics of the user population are driven by the interplay between continuous engagement (viewing) and discrete resampling events (swiping). We first describe the continuous phase.

Controlled Attention Process. Consider a continuum of ex-ante homogeneous users indexed by $i \in [0, N]$. Let $Z_t^i \in \mathbb{R}_+$ denote the instantaneous attention level (or physiological arousal) of user i at time t during the viewing of a single video. We model Z_t^i as a controlled Geometric Brownian Motion (GBM):

$$\mathrm{d}Z_t^i = \left(\mu + a(t, Z_t^i)\right) Z_t^i \mathrm{d}t + \sigma Z_t^i \mathrm{d}W_t^i, \tag{1}$$

where W_t^i are independent standard Brownian motions capturing the intrinsic volatility of content stimuli.

- **Natural Decay ($\mu < 0$):** The parameter μ represents the baseline rate of attention decay (i.e., the onset of boredom). In the absence of stimuli, user attention naturally drifts toward zero, reflecting the entropy of the swiping mechanism.
- **Algorithmic Modulation ($a(t, z)$):** The control variable $a(t, z) \in [a_{\min}, a_{\max}]$ (where $a_{\min} < 0$) represents the platform's *intervention intensity*. Unlike traditional editorial editing, this corresponds to bidirectional *Sensory Modulation*:
  - **Positive Stimulation ($a > 0$):** Using real-time features (e.g., visual hooks, intense audio cues) to artificially inject a positive drift that *counteracts* natural decay (Retention).

– **Negative Nudging ($a < 0$):** Strategically reducing stimuli or withholding gratification to *accelerate* decay. This allows the algorithm to proactively "flush" users out of low-yield states to trigger a new swipe (Impression), serving as a micro-foundation for the Algorithmic Liquidity Trap.

We denote the effective drift as $\tilde{\mu}(t, z) = \mu + a(t, z)$. This SDE governs the user's path *only* until an endogenous stopping time τ^*. Upon hitting the churn boundary $b^*(t)$, the user exits the current path and triggers a "swipe" event, governed by the traffic recycling mechanism.

Modeling Remark: From Agents to Fields. In defining the macroscopic evolution, we adopt a "top-down" Mean-Field approach. While methods like heterogeneous-agent models (Aiyagari, 1995) are valuable in macroeconomics, they often rely on static equilibrium concepts. Given our focus on the *velocity* of user flows and the *recycling* dynamics, we model the aggregate state directly via a Mean-Field PDE. Crucially, our ecosystem is an *open system with internal recycling*, governed by a *Non-Local Fokker-Planck Equation*. This structure allows us to capture the complex feedback between the churn boundary b^* and the re-entry distribution $g(z)$—a necessary feature to model the "Algorithmic Liquidity Trap."

3.3 The User's Problem: Optimal Stopping with Endogenous Thresholds

Given the controlled attention dynamics described in Equation (1), the user solves an optimal stopping problem to determine the precise moment to "swipe" away from the current video.

Value Function and the HJB Equation. The user's objective is to maximize total discounted utility. The value function $v(t, z)$ is defined as the supremum over all admissible stopping times τ:

$$v(t, z) = \sup_{t \le \tau < \infty} \mathbb{E}\left[\int_t^\tau e^{-\rho(s-t)} u_{util}(Z_s) \mathrm{d}s + e^{-\rho(\tau - t)} \left(K(\tau, u(\tau)) - C \right) \Big| Z_t = z \right]. \tag{2}$$

We employ a CRRA utility function $u_{util}(z) = \frac{z^{1-\gamma}}{1-\gamma}$ (with $\gamma > 1$). The term C represents the cognitive switching cost.

Crucially, the payoff upon switching, $K(t, u)$, is not fixed but depends on the platform's concurrent *Matching Control* $u(t)$. This constitutes a *Variational Inequality (VI)* for the value function:

$$\min \left\{ \rho v - \frac{\partial v}{\partial t} - \mathcal{L}^a v - u_{util}(z), v(t, z) - \big(K(t, u(t)) - C \big) \right\} = 0, \tag{3}$$

where $\mathcal{L}^a$ is the generator of the controlled diffusion:

$$\mathcal{L}^a v = (\mu + a(t,z))z\frac{\partial v}{\partial z} + \frac{1}{2}\sigma^2 z^2 \frac{\partial^2 v}{\partial z^2}. \tag{4}$$

The *optimal stopping boundary* (or Swiping Threshold) $b^*(t)$ is the free boundary separating the continuation region from the stopping region:

$$b^*(t) = \inf\left\{z \in \Omega \mid v(t,z) > K(t,u(t)) - C\right\}. \tag{5}$$

Equivalently, under the threshold structure, $b^*(t)$ is characterized by the *value-matching* condition

$$v\big(t, b^*(t)\big) = K\big(t, u(t)\big) - C. \tag{6}$$

Users with attention $z \le b^*(t)$ immediately execute a "swipe".

Boundary Conditions (Value-Matching and Smooth-Fit). Since the switching payoff $K(t,u(t)) - C$ is constant in z, the standard smooth-pasting condition implies

$$\frac{\partial v}{\partial z}\big(t, b^*(t)\big) = 0. \tag{7}$$

At the artificial truncation boundaries $z_{\min}, z_{\max}$ (used in our numerical solver), we impose reflecting (Neumann) conditions consistent with the conservative fixed-domain scheme:

$$\frac{\partial v}{\partial z}(t, z_{\min}) = 0, \qquad \frac{\partial v}{\partial z}(t, z_{\max}) = 0, \tag{8}$$

together with a standard growth/transversality condition ensuring finiteness of v under discounting. *Remark (Numerical truncation):* these boundary conditions are imposed on the truncated computational domain for numerical stability; in our experiments we choose $z_{\max}$ sufficiently large (and $z_{\min}$ sufficiently small) and confirm that the computed policy and threshold are insensitive to further enlarging the domain.

The Endogenous Outside Option (Coupling Variable). The variable $K(t,u)$, which serves as the boundary condition for the user's problem, represents the expected value of the "next video". In our model, this creates a *Fixed-Point problem*: the value of the outside option depends on the value function itself. Specifically, K is determined by the platform's investment $u(t)$, which shapes the content quality distribution $g(z;u)$:

$$K(t,u(t)) = \int_{z_{\min}}^{z_{\max}} v(t,z')g(z';u(t))\mathrm{d}z'. \tag{9}$$

Remark (Implicit Fixed-Point Structure). Since $v(t,z') = K(t,u(t)) - C$ for all $z' \le b^*(t)$ (by value-matching), Equation (9) is an *implicit* equation in K. Letting $G(t) \equiv \int_{z_{\min}}^{b^*(t)} g(z';u)\,\mathrm{d}z'$ denote the probability that the next video lands in the stopping region, one can verify that $K = G(K-C) + \int_{b^*(t)}^{z_{\max}} v\,g\,\mathrm{d}z'$, i.e., $K = \frac{-G\,C + \int_{b^*}^{z_{\max}} v\,g\,\mathrm{d}z'}{1-G}$. In our numerical solver we treat this as a fixed-point iteration: given a candidate K, we solve the QVI for v, update K via (9), and iterate to convergence.

This equation highlights the *Dopamine Treadmill* effect: An increase in platform matching investment $u(t)$ shifts $g(z;u)$ towards higher quality. This raises the expected switching value K, which in turn raises the user's stopping threshold $b^*(t)$. Paradoxically, better recommendation algorithms may make users *more* impatient, as the opportunity cost of watching mediocre content increases. This mechanism mathematically drives the acceleration of swiping velocity, serving as the micro-foundation for the "High-Velocity Saturation" regime observed in our results.

Theoretical Well-posedness

Before characterizing the equilibrium, we establish the theoretical well-posedness of the user's sub-problem defined by the QVI in (3).

Proposition 1 (Well-posedness of the User QVI)**.** *Given a fixed platform control policy $\{a(t,z),u(t)\}$ and a fixed outside option process $K(t,u(t))$, the QVI in (3) admits a unique viscosity solution $v(t,z)$ under standard boundedness and discounting conditions (Fleming & Soner, 2006).*

Moreover, suppose the re-entry distribution $g(\cdot;u(t))$ is a nonnegative density on $[z_{\min}, z_{\max}]$ with $\int g\,\mathrm{d}z = 1$ and $\|g\|_\infty < \infty$, and we work on the truncated state domain with discounting $\rho > 0$ so that v is finite. Then the coupling (9) defines a well-posed fixed-point problem in K. In our numerical implementation we solve it via a monotone (relaxed) value-iteration scheme and verify convergence of K in sup norm.

Proposition 2 (Threshold Structure)**.** *For the CRRA utility function with $\gamma > 1$, the value function $v(t,z)$ is monotonically increasing in z. Consequently, the optimal stopping region is connected and characterized by a unique threshold $b^*(t)$:*

$$\mathcal{S}_t = [z_{\min}, b^*(t)]. \tag{10}$$

Proof. The result follows from the monotonicity of the generator for one-dimensional diffusions. Detailed proof is provided in Appendix A.1.

This structural property is computationally significant: it allows us to search for a single scalar $b^*(t)$ (e.g., via bisection) rather than evaluating the stopping decision for every grid point, reducing the complexity from $O(N_z)$ to $O(\log N_z)$.

3.4 Baseline Analysis: The "Organic" Equilibrium

Before solving the full Stackelberg game with active algorithmic intervention, we simulate the user subgame under a neutral policy ($a = 0, u = u_0$). This analysis is critical not only to validate structural stability but to establish an *"Organic Benchmark"*—representing the natural state of user attention before it is distorted by the Algorithmic Liquidity Trap.

To ensure empirical relevance, we employ parameters calibrated from the KuaiRand dataset (Section 4). The equilibrium is computed using the *Fully Implicit Finite Difference*

Solver (Algorithm 1 in Appendix).

3.4.1 Equilibrium Properties: The Healthy Baseline

Figure 1 presents the numerical solution. The results confirm that the system converges to a unique *Stationary Mean-Field Equilibrium (MFE)*.

Stationary Strategy and Patience (Left Panel). The optimal stopping boundary $b^*(t)$ converges to a constant threshold $b^*_\infty \approx 0.65$. The strict positivity of this threshold ($b^* > z_{\min}$) confirms that rational users exercise their "option to switch" strategically.

Logistic Growth and Flux Conservation (Middle Panel). The macroscopic dynamics reveal two structural insights that contrast sharply with the high-velocity regime discussed later:

- **Open System Dynamics:** The total active population $N(t)$ stabilizes at $N^* \approx 83.68$. This is constrained by the natural churn rate ϵ, reflecting the platform's "carrying capacity" under organic conditions.
- **Validating the "Healthy Heartbeat":** The swipe flux $\Phi(t)$ converges to a constant rate $\Phi^* \approx 1.36$ swipes/sec. Crucially, this implies an *average dwell time* of[1]:

$$\tau_{base} = \frac{N^*}{\Phi^*} \approx \frac{83.68}{1.36} \approx 61.53 \text{ seconds.} \tag{11}$$

  Unlike the "Phantom Liquidity" regime identified under algorithmic maximization—where dwell times collapse to a physiological floor well below the IAB/MRC standard of > 2s—this baseline flux represents *High-Quality Inventory* that comfortably satisfies viewability standards (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). It demonstrates that the "physiological cliff" is not an inherent property of the user, but an induced state by the algorithm.

Distributional Convergence (Right Panel). The heatmap confirms the formation of a stationary distribution $f_\infty(z)$. The high-density "engagement core" (yellow region) shows that without aggressive intervention, user attention naturally settles into a sustainable rhythm of consumption and switching.

3.4.2 Comparative Statics: Mechanisms of Sensitivity

To probe the sensitivity of the equilibrium, we conduct comparative statics along two dimensions: algorithmic retention intensity (μ) and content ecosystem quality (μ_g). The results, illustrated in Figure 2, reveal distinct mechanisms governing ecosystem health.

[1]This relationship is derived from *Little's Law* (Little et al., 1961) in queueing theory, which states that the long-term average number of customers in a stationary system (N^*) is equal to the long-term average effective arrival rate (Φ^*) multiplied by the average time a customer spends in the system (τ).

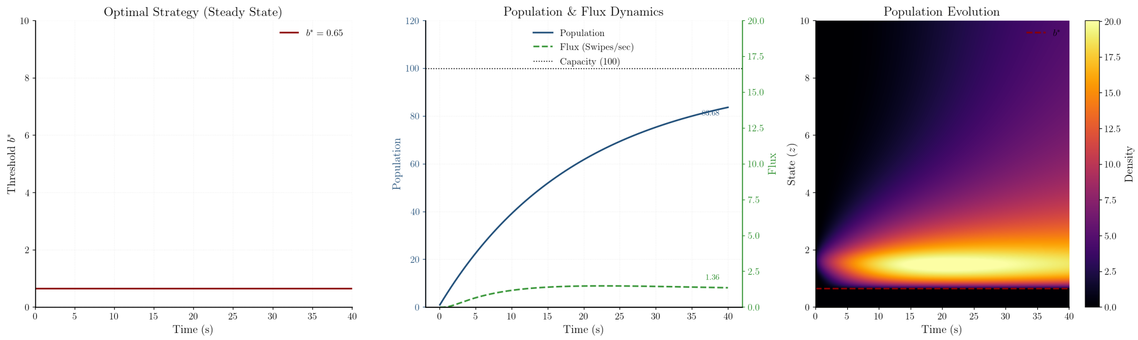

Figure 1: **Baseline Organic Equilibrium.** Under neutral policies, the system stabilizes at a healthy dwell time of $\tau \approx 61.5\,\text{s}$. (Left) Stationary stopping boundary. (Middle) Logistic growth of population and stable flux. (Right) Stationary distribution of attention.

Experiment A: Algorithmic Intensity and The Retention Dividend. We vary the drift parameter μ to simulate "Low Stickiness" ($\mu = -0.15$), "Baseline" ($\mu = -0.0235$), and "High Stickiness" ($\mu = 0.2$).

- **The Lock-in Effect (Top-Left):** Strong positive reinforcement ($\mu = 0.2$) *lowers* the optimal stopping threshold $b^* \approx 0.63$ (compared to Baseline 0.65). Users become "locked in," as the enhanced immediate gratification dominates the option value of switching.

- **The Retention Dividend (Bottom-Left):** We find that high stickiness ($\mu = 0.2$) results in the *highest* steady-state population ($N \approx 86.6$). *Mechanism:* The ecosystem dynamics are governed by the balance between acquisition and churn. By increasing retention ($\mu = 0.2$), the platform effectively slows down the swipe flux. Since user churn is coupled to swiping events (churn rate $\propto \epsilon\Phi$), a slower velocity preserves the user stock. Conversely, the "Low Stickiness" environment ($\mu = -0.15$) accelerates swiping but drains the population ($N \approx 72.9$) due to excessive churn exposure. This confirms that within organic limits, retention acts as a buffer against population decay.

Experiment B: Content Ecosystem Maturity. We shift the mean content quality μ_g to simulate "Low Quality" (1.3), "Baseline" (1.65), and "High Quality" (2.0) ecosystems.

- **The Dopamine Treadmill (Top-Right):** Improving the content pool ($\mu_g = 2.0$) strictly *raises* the reservation utility ($b^* \approx 0.73$, up from Baseline 0.68). *Economic Interpretation:* This reflects an *"Inflation of Expectations."* As the environment improves, users rationally internalize the higher expected value of the "next video" (K). They become more selective, requiring a higher utility threshold to justify staying on the current item.

- **Active Searching (Bottom-Right):** Consequently, the High Quality ecosystem generates the *highest* traffic flux ($\Phi \approx 1.43$). *Mechanism:* High quality encourages *Active*

Searching. Paradoxically, better content does not simply make users "sit back"; it encourages them to swipe more frequently to maximize utility per unit of time (1.43 swipes/sec vs. 1.30 in Low Quality). This mechanism explains why purely quality-driven algorithms can inadvertently increase user impatience, as the opportunity cost of watching any single video increases. Interestingly, we observe that the steady-state population remains *inelastic* to content quality ($N^* \approx 84$ across all regimes). This suggests that improving recommendation quality essentially accelerates the ecosystem's "metabolism"—increasing the frequency of interactions (Flux)—without significantly altering the platform's carrying capacity.

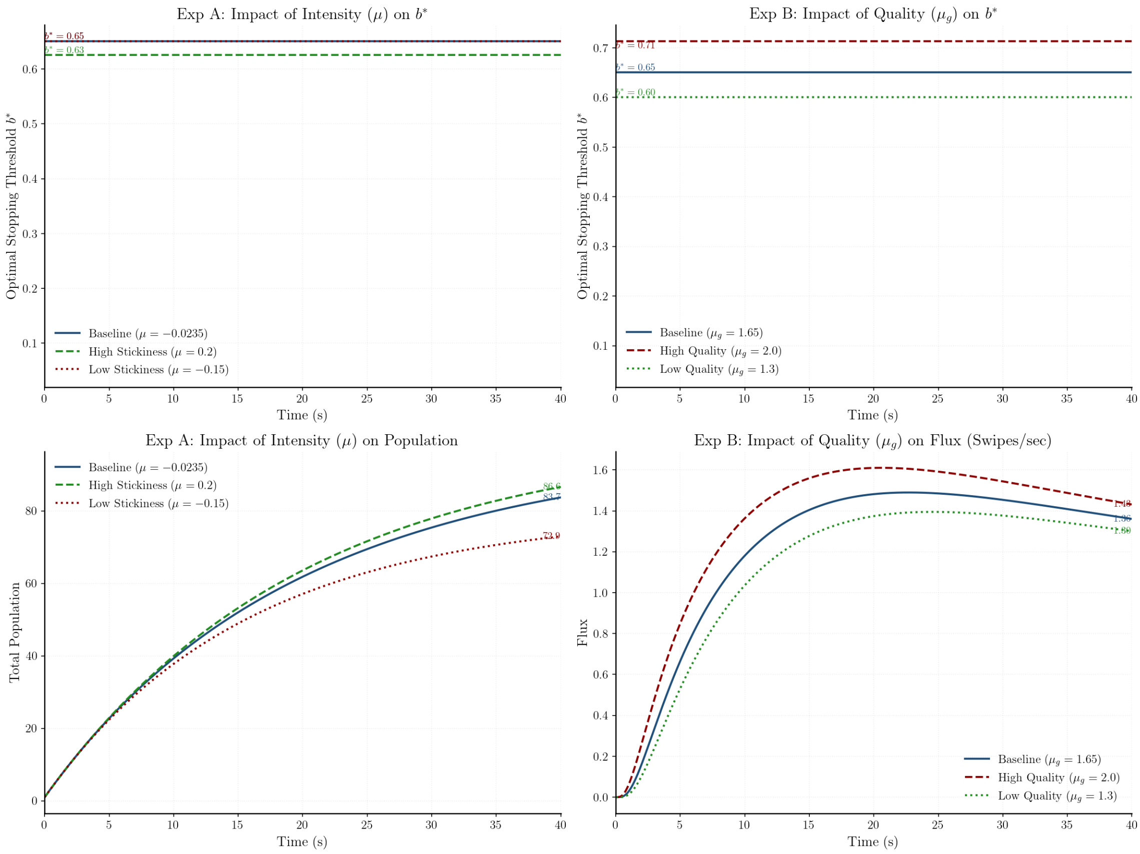

Figure 2: **Comparative Statics. (Left Column):** High stickiness ($\mu = 0.2$) generates a **Retention Dividend**, significantly boosting population ($N \approx 86.6$) while reducing flux to near-zero. **(Right Column):** Improving content quality ($\mu_g = 2.0$) accelerates the ecosystem metabolism. Note that we report Flux (Φ) for quality sensitivity because the population level (N) remains largely invariant to quality changes ($\Delta N < 1\%$), indicating that better content drives activity velocity rather than market scale.

3.5 The Macro-Dynamics: Non-Local Fokker-Planck Equation

As the population size $N \to \infty$, the aggregate behavior of users is described by the population density $f(t, z)$ (an *unnormalized* measure density). The evolution of $f(t, z)$ follows mass balance, adjusted for inflows (acquisition and recycling) and outflows (true churn).

Derivation of the Non-Local PDE. Standard arguments involving Itô's formula and integration by parts (see Gardiner et al., 2004) yield the differential operator $\mathcal{L}^*$ adjoint to the generator $\mathcal{L}^a$. However, unlike standard Fokker-Planck equations that assume mass conservation, our ecosystem acts as an open system with internal loops.

Physical domain and absorption. Conceptually, the density evolves on the time-dependent physical domain $\Omega_t = [b^*(t), z_{\max}]$ (users who hit $b^*(t)$ stop watching the current video). On Ω_t we impose the absorbing boundary condition $f(t, b^*(t)) = 0$ and a reflecting boundary at $z_{\max}$.

Non-local sources and units. The re-entry kernel $g(\cdot; u(t))$ is a probability density on $[z_{\min}, z_{\max}]$ with $\int g(z; u)\, dz = 1$ (units $1/z$). Therefore, $(1-\epsilon)\Phi_{swipe}(t)g(z; u(t))$ has units "users/(time$\cdot z$)", matching $\partial_t f$.

Under this physical interpretation, the density satisfies:

$$\frac{\partial f}{\partial t}(t,z) = \underbrace{\mathcal{L}^* f(t,z)}_{\text{Gaze Dynamics}} + \underbrace{A(t)\delta(z-z_p)}_{\text{External Acquisition}} + \underbrace{(1-\epsilon)\Phi_{swipe}(t)g(z;u(t))}_{\text{Traffic Recycling}}, \qquad z \in \Omega_t. \tag{12}$$

Mass balance implied by absorption. Integrating (12) over Ω_t and using the reflecting boundary at $z_{\max}$ yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} f(t,z)\, \mathrm{d}z = A(t) - \epsilon\, \Phi_{swipe}(t), \tag{13}$$

where the loss term is the non-recycled fraction ϵ of the swipe flux (true churn).

Fixed-domain implementation (cut). For numerical stability, we implement a conservative fixed-domain scheme on $\Omega = [z_{\min}, z_{\max}]$ and enforce the absorption at $b^*(t)$ via a projection (cut) after each implicit PDE step; this cut removes exactly the swipe mass, recycles a fraction $(1-\epsilon)$ via g, and loses a fraction ϵ, matching (13) discretely.

where the adjoint operator is:

$$\mathcal{L}^* f = -\frac{\partial}{\partial z}\left[(\mu + a(t,z))zf\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma^2 z^2 f\right]. \tag{14}$$

We denote the associated probability current by

$$J(t,z) \triangleq (\mu + a(t,z))zf(t,z) - \frac{1}{2}\frac{\partial}{\partial z}\left(\sigma^2 z^2 f(t,z)\right). \tag{15}$$

Interpretation of Source Terms. Equation (12) distinguishes our model from standard MFG frameworks through two distinct source mechanisms:

1. **External Acquisition (Dirac Source):** The term $A(t)\delta(z-z_p)$ represents the injection of new users. The control variable $A(t)$ corresponds to marketing spend (akin to the Bass, 1969 diffusion model), while the Dirac delta $\delta(z-z_p)$ signifies a standardized onboarding process.

2. **Traffic Recycling (The Feedback Loop):** The term $(1-\epsilon)\Phi_{swipe}(t)g(z;u(t))$ captures

the closed-loop nature of short-video platforms. When a user swipes, a fraction $(1 - \epsilon)$ immediately re-enters the cycle. *Theoretical Note:* This non-local coupling is the mathematical root of the *"Algorithmic Liquidity Trap."* Since the platform can boost population density $f(t,z)$ by accelerating the recycling flux $\Phi_{swipe}(t)$, unconstrained optimization algorithms may aggressively drive users to swipe (increasing Φ) rather than extending their gaze (retention), inadvertently creating a high-velocity, low-value ecosystem.

Boundary Conditions and Flux Dynamics. To align with our conservative "solver-in-the-loop" implementation on a fixed computational container, we evolve $f(t,z)$ on the static domain $\Omega = [z_{\min}, z_{\max}]$ subject to *zero-flux (reflecting)* boundary conditions at both ends:

$$f(0,z) = f_0(z), \quad \text{(Initial Distribution)} \tag{16}$$

$$J(t, z_{\min}) = 0, \qquad J(t, z_{\max}) = 0. \quad \text{(Zero-Flux Boundaries)} \tag{17}$$

The reflecting boundaries prevent artificial mass leakage through the computational container during each implicit time step.

The *absorbing* nature of the endogenous stopping threshold $b^*(t)$ is then enforced via a *projection (cut)* condition applied after each PDE update:

$$f(t,z) = 0, \qquad \forall z < b^*(t), \tag{18}$$

so that any mass that has entered the stopping region is strictly removed and counted as swipe flux.

$$\Phi_{swipe}(t_n) \triangleq \frac{1}{\Delta t} \int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z)\, \mathrm{d}z. \tag{19}$$

where $f^*(t_n, \cdot)$ denotes the post-implicit (pre-cut) density in our conservative time-stepping scheme. This discrete integral formulation is numerically robust and guarantees exact consistency between the computed swipe flux and the mass removed by the projection step.

In the smooth continuous-time limit, one may relate the swipe flux to the probability current at the threshold via $\Phi_{swipe}(t) \approx -J(t, b^*(t))$. *Important clarification:* throughout the paper, our *numerical* swipe flux is computed by the conservative cut-based mass integration in Eq. (19) (mass removed by the projection step divided by Δt). The current-based identity is only a continuous-limit heuristic; with a time-varying free boundary $b^*(t)$, an exact flux-current equality generally requires additional moving-boundary correction terms.

This flux $\Phi_{swipe}(t)$ is the coupling variable that closes the feedback loop. In the platform's optimization problem (defined next), this flux represents the *supply of ad inventory*. The conflict between maximizing this inventory supply (Φ) and maintaining user retention (τ) constitutes the central tension of our model.

Remark 1 (Mathematical Well-posedness). The inclusion of source terms in Fokker-Planck equations is well-grounded. Zhu (2022) establishes the existence of weak solutions for kinetic FPEs with source terms, showing that the solution norm depends on the source norm. In the context of MFG, Claisse, Ren, and Tan (2023) derive similar equations where state-dependent birth/death rates lead to solutions that are finite measures rather than probability measures—a property that perfectly matches our non-conservative user population dynamics.

Remark 2 (Numerical Implementation). While the Dirac delta $\delta(z - z_p)$ is theoretically convenient, for numerical implementation (Section 4), we approximate it using a smooth Gaussian kernel to prevent Gibbs phenomena and ensure spectral stability.

Remark 3 (Distinction from Closed Queueing Networks). It is worth noting that the "Traffic Recycling" topology bears a structural resemblance to Closed Queueing Networks (CQN) or Gordon-Newell networks (Gordon & Newell, 1967). However, our framework differs fundamentally in its *control objective*. In classical CQN, the service mechanism is typically passive, and the objective is often to manage congestion or maximize throughput given fixed service requirements. In our Short-Video Economy, the "service time" (dwell time) is *strategically endogenous*. The platform (server) actively intervenes via $a(t, z)$ to manipulate the rate of recycling. The innovation lies not just in the closed loop itself, but in the *Stackelberg gaming* of this loop: the platform optimizes the *velocity* of circulation against the user's physiological patience, creating the unique "Algorithmic Liquidity Trap" phenomenon that standard queueing models do not capture.

3.6 The Leader's Problem: Mean-Field Control with Dual Monetization

In the standard Mean-Field Game literature, the principal often maximizes a functional of the population state alone (e.g., total density). However, as identified in the audit of short-video economics, a model focusing solely on total attention time ($\int zf$) mimics a subscription-based model (like Netflix) but fails to capture the ad-driven revenue model of platforms like TikTok or Reels. In these ecosystems, revenue is generated not just by "gazing" (retention) but significantly by "swiping" (impression generation).

To resolve the "Stalemate of Perfection" paradox—where optimal retention implies zero ad inventory—we reconstruct the platform's objective and control mechanisms. The platform acts as a Stackelberg leader, optimizing a vector of controls to balance retention intensity against the need for traffic circulation (swiping).

3.6.1 Reformulation: Decoupling Scale and Structure

To explicitly separate the economic effects of market growth (Scale) from user engagement quality (Structure), we perform a change of variables on the state space. We decompose the unnormalized population density $f(t, z) \in \mathcal{M}^+(\Omega)$ into two orthogonal components: the total population mass $N(t) \in \mathbb{R}_+$ and the probability density function $m(t, z) \in \mathcal{P}(\Omega)$.

The relationship is defined as:

$$\begin{aligned} f(t,z) &= N(t)\cdot m(t,z), \\ \text{where } N(t) &= \int_\Omega f(t,z)\,\mathrm{d}z, \\ \text{and } \int_\Omega m(t,z)\,\mathrm{d}z &= 1. \end{aligned} \tag{20}$$

Consequently, the platform's value functional is redefined as $\mathcal{W}(f) \equiv V(N,m)$. This decomposition allows us to distinguish the platform's two strategic imperatives: managing the *market size* (via acquisition A and churn ϵ) and optimizing the *attention quality* (via retention a and matching u).

3.6.2 The Reconstructed Objective Functional and Constraints

The platform seeks to maximize the total discounted net profit over an infinite horizon. The objective functional $J(a,u,A)$, expressed in terms of the decomposed variables, combines retention value, impression revenue, and operational costs:

$$\begin{aligned} \max_{a,u,A} J = \int_0^\infty e^{-rt} \Bigg[& \underbrace{N(t)\int_{b^*(t)}^{z_{\max}} \left(\lambda_1 z - \frac{\xi_1}{2}a^2\right) m\,\mathrm{d}z}_{\text{Net Retention Value}} \\ & + \underbrace{N(t)\lambda_2\phi_{swipe}(m) - \frac{\xi_2}{2}u(t)^2}_{\text{Net Impression Value}} \\ & - \underbrace{\Psi(A(t))}_{\text{Acq. Cost}} \Bigg] \mathrm{d}t \end{aligned} \tag{21}$$

Economic Interpretation of the Objective: Equation (21) encapsulates the fundamental tension of the attention economy:

- **The Editor's Incentive ($\lambda_1 z$):** Rewards prolonged engagement depth (Duration).
- **The Dealer's Incentive ($\lambda_2\phi_{swipe}$):** Rewards the velocity of card dealing (Flux).

The conflict arises because ϕ_{swipe} is physically generated by terminating the viewing process. If λ_2 (Ad Revenue) dominates, the algorithm is mathematically incentivized to accelerate the "churn" from individual videos to maximize aggregate inventory, potentially creating the "Algorithmic Liquidity Trap."
Subject to the following constraints:

1. **Scale Dynamics (ODE):** The evolution of the total population $N(t)$ is governed by the balance between acquisition and net churn:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = A(t) - \epsilon N(t)\phi_{swipe}(m). \tag{22}$$

2. **Structural Dynamics (PDE):** The probability density $m(t, z)$ evolves according to the Kolmogorov equation with dilution and recycling terms, defined on $z \in (b^*(t), z_{\max}]$:

$$\begin{aligned}\frac{\partial m}{\partial t} &= \mathcal{L}^*[a]m \\ &+ \frac{A(t)}{N(t)}(\delta_{z_p} - m) \\ &+ (1-\epsilon)\phi_{swipe}(m)\, g(z; u(t)) + \epsilon\, \phi_{swipe}(m)\, m.\end{aligned} \tag{23}$$

Derivation from $f = Nm$ (term-by-term consistency). Start from the unnormalized equation $\partial_t f = \mathcal{L}^*[a]f + A(t)\delta_{z_p} + (1-\epsilon)\Phi_{swipe}(t)g(z; u(t))$ with $\Phi_{swipe} = N\phi_{swipe}(m)$. Writing $f = Nm$ gives $\partial_t f = \dot{N}\, m + N\partial_t m$ and $\mathcal{L}^*[a]f = N\mathcal{L}^*[a]m$. Divide by N to obtain

$$\partial_t m = \mathcal{L}^*[a]m + \frac{A}{N}\delta_{z_p} + (1-\epsilon)\phi_{swipe}(m)g - \frac{\dot{N}}{N}m.$$

Using the scale ODE $\dot{N} = A - \epsilon N\phi_{swipe}(m)$, the dilution/renormalization term becomes $-\frac{\dot{N}}{N}m = -\frac{A}{N}m + \epsilon\phi_{swipe}(m)m$, yielding exactly the PDE above.

Normalization check (absorption sign convention). Write the Fokker–Planck equation in conservative form $\partial_t m = -\partial_z J_m + \text{(sources)}$ with current

$$J_m(t, z) \equiv (\mu + a(t, z))z\, m(t, z) - \frac{1}{2}\partial_z\big(\sigma^2 z^2 m(t, z)\big).$$

On the physical domain $\Omega_t = [b^*(t), z_{\max}]$, the absorbing boundary means $m(t, b^*(t)) = 0$ and the flux through $b^*(t)$ is

$$\phi_{swipe}(m) \equiv -J_m\big(t, b^*(t)^+\big) \geq 0,$$

i.e., the outward normal at the left boundary points toward decreasing z, so a *negative* current J_m corresponds to a *positive* outflow. With a reflecting (zero-flux) condition at $z_{\max}$, integrating gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} m(t, z)\, \mathrm{d}z = -\phi_{swipe}(m) \quad \text{(before adding recycling/renormalization)}.$$

The last line of the PDE reinjects total mass $\phi_{swipe}(m)$ (recycle share $(1-\epsilon)$ drawn from g plus the renormalization term ϵm induced by true churn), ensuring $\int m\, \mathrm{d}z \equiv 1$.

Consistency with cut-based Φ_{swipe}. In the fixed-domain numerical scheme we define the total swipe flux by mass removed by the projection step, $\Phi_{swipe}(t_n) = \frac{1}{\Delta t}\int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z)\, \mathrm{d}z$ (Eq. (19)). When $b^*(t)$ is time-independent (or in the $\Delta t \to 0$ smooth limit under a fixed boundary), this converges to the boundary-current identity $\Phi_{swipe}(t) = N(t)\, \phi_{swipe}(m) = -N(t)\, J_m(t, b^*(t)^+)$. For a moving boundary $b^*(t)$, the cut-based definition remains the exact accounting identity used in the paper, while the gradient/current formula should be interpreted as a continuous-limit heuristic (and may require a moving-boundary correction term).

3. **Flux Consistency:** The per-capita swipe flux admits the standard boundary-flux representation in the smooth fixed-boundary limit:

$$\phi_{swipe}(m) \triangleq \frac{1}{2}\sigma^2(b^*(t))^2 \frac{\partial m}{\partial z}(t, b^*(t)). \tag{24}$$

   The corresponding *total* swipe flux satisfies $\Phi_{swipe}(t) = N(t)\, \phi_{swipe}(m)$.

   Important clarification (moving boundary & numerical definition). Equation (24) should be interpreted as a continuous-limit *heuristic* that relies on sufficient regularity of m at the absorbing boundary and is exact when the boundary is fixed (or when moving-boundary correction terms are negligible). Throughout the paper (and in our computations), the *defining* quantity is the cut-based accounting identity $\Phi_{swipe}(t_n) = \frac{1}{\Delta t}\int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z)\, \mathrm{d}z$ (Eq. (19)), which exactly equals the mass removed by the projection step divided by Δt. When $\Delta t \to 0$ under a fixed boundary, this converges to the boundary current/gradient representation; for a time-varying free boundary $b^*(t)$, the cut-based definition remains the correct conservation-consistent notion of swipe flux used to close the feedback loop.

4. **User Incentive Compatibility:** The stopping boundary $b^*(t)$ arises endogenously from the users' optimal stopping problem (smooth-pasting condition):

$$b^*(t) = \inf \{z \in \Omega \mid v(t, z) > K(t, u(t)) - C\}\,. \tag{25}$$

3.6.3 The Master Equation (Explicit Form)

The platform's value function $V(N, m)$ satisfies the infinite-dimensional Hamilton-Jacobi-Bellman equation. By solving the maximization problem using the structure-scale decomposition (see Appendix C), we derive the explicit equation:

$$\begin{aligned} rV(N,m) = \sup_{a,u,A} \Bigg\{ & \underbrace{\pi(N,m,a,u,A)}_{\text{Flow Payoff}} + \frac{\partial V}{\partial N}\Big(A - \epsilon N \phi_{\text{swipe}}(m)\Big) \\ & + \int_\Omega \frac{\delta V}{\delta m}(z)\Bigg[\mathcal{L}^*[a]m + \frac{A}{N}(\delta_{z_p} - m) \\ & + (1-\epsilon)\phi_{\text{swipe}}(m)\, g(z;u) + \epsilon\, \phi_{\text{swipe}}(m)\, m\Bigg]\,\mathrm{d}z \Bigg\}. \end{aligned} \tag{26}$$

Expanding the optimal controls yields the non-linear form. Note that the boundary term arising from $\mathcal{L}^*$ is implicitly contained within the Hamiltonian structure:

$$\begin{aligned} rV(N,m) = N \int \Big[&\lambda_1 z + \mu z \partial_z \lambda_m \\ &+ \frac{1}{2}\sigma^2 z^2 \partial_{zz}\lambda_m \Big] m \, \mathrm{d}z \\ &+ \underbrace{N \int \frac{z^2}{2\xi_1} \big(\partial_z \lambda_m\big)^2 m \, \mathrm{d}z}_{\text{Quadratic Retention Gain}} \\ &+ H_{acq}\left(\frac{\partial V}{\partial N} + \frac{1}{N}\big(\lambda_m(z_p) - \bar{\lambda}_m\big) \right) \\ &+ \mathcal{H}_{match}\big(N\phi_{swipe}(m), \lambda_m\big), \end{aligned} \tag{27}$$

where $\lambda_m(z) \equiv \frac{1}{N}\frac{\delta V}{\delta m}(z)$ represents the normalized structural shadow price, and $\bar{\lambda}_m = \int \lambda_m m \, \mathrm{d}z$. The functions H_{acq} and $\mathcal{H}_{match}$ denote the Legendre-Fenchel transforms (convex conjugates) of the cost functions associated with acquisition A and matching control u, respectively. Specifically, $H_{acq}(Y) = \sup_{A \geq 0} \{Y \cdot A - C_{acq}(A)\}$, where the argument Y represents the net shadow value of a new user.

3.6.4 Optimal Strategic Responses

The First-Order Conditions (FOCs) describe the optimal strategy:

- **Optimal Retention Intensity:**

$$a^*(t,z) = \frac{z}{\xi_1}\frac{\partial}{\partial z}\left(\frac{1}{N}\frac{\delta V}{\delta m}(z)\right). \tag{28}$$

  The platform applies friction proportional to the gradient of the structural shadow price, fighting boredom where the marginal gain of engagement quality is highest.

- **Optimal Matching Investment:**

$$\begin{aligned} \xi_2 u^*(t) = (1-\epsilon) &N(t)\phi_{swipe}(m) \\ &\cdot \int \frac{1}{N}\frac{\delta V}{\delta m}(z)\frac{\partial g}{\partial u}(z; u^*) \, \mathrm{d}z. \end{aligned} \tag{29}$$

  Investment is proportional to traffic flux $N\phi_{swipe}$. This creates a *positive feedback loop*: higher flux justifies better algorithms, and better algorithms (via the Dopamine Treadmill) further accelerate flux. This self-reinforcing mechanism explains the system's tendency to overheat into the "Phantom Liquidity" regime.

- **Optimal Acquisition:**

$$\Psi'(A^*) = \frac{\partial V}{\partial N} + \frac{1}{N}\left(\frac{\delta V}{\delta m}(z_p) - \int \frac{\delta V}{\delta m} m \, \mathrm{d}z\right). \tag{30}$$

Marketing spend balances the value of adding a user ($\partial_N V$) against the dilution effect of adding a user at the specific entry state z_p.

3.6.5 Mean Field Equilibrium

Substituting the optimal controls $a^*(t,z)$, $u^*(t)$, and $A^*(t)$ derived in Equations (28)–(30) back into the structural dynamics, we obtain the fully coupled system governing the platform ecosystem.

Definition 1 (Mean-Field Stackelberg Equilibrium)**.** A Mean-Field Equilibrium is a tuple of time-dependent functions $(V, m, \Phi_{swipe}, b^*)$ such that:

1. **Optimality:** Given the population distribution m, the value function V satisfies the Master HJB Equation (26), and the controls $\{a^*, u^*, A^*\}$ maximize the Hamiltonian.

2. **Consistency:** Given the optimal controls, the population density m evolves according to the controlled Non-Local Fokker-Planck Equation:

$$\begin{cases} \dfrac{\partial m}{\partial t} = -\dfrac{\partial}{\partial z}\Big((\mu + a^*)zm\Big) \\ \qquad\qquad + \dfrac{1}{2}\sigma^2 \dfrac{\partial^2}{\partial z^2}(z^2 m) + \mathcal{S}(m, V), \\ \\ rV = \sup\limits_{a,u,A} \mathcal{H}\left(N, m, V, \dfrac{\delta V}{\delta m}, a, u, A\right), \end{cases} \tag{31}$$

   where $\mathcal{S}(m, V)$ represents the source terms driven by acquisition $A^*[V]$ and recycled traffic flux $\Phi_{swipe}(m)$; $\mathcal{H}$ denotes the *pre-optimized* Hamiltonian functional defined in Eq. (26).

3. **Boundary Consistency:** The swipe flux $\Phi_{swipe}(t)$ and the free boundary $b^*(t)$ satisfy the smooth-pasting condition generated by the users' optimal stopping problem:

$$b^*(t) = \inf\{z \in \Omega \mid v(t,z) > K(t, u^*(t)) - C\}. \tag{32}$$

Due to the forward-backward structure (forward in time for m, backward for V) and the non-local coupling via Φ_{swipe}, this system does not admit a closed-form analytical solution. In the following section, we propose a computational framework for solving the stationary equilibrium.

4 Structure Calibration and Solution Algorithm

To bridge the gap between our theoretical Mean-Field Stackelberg framework and the empirical reality of short-video platforms, we adopt a two-stage methodological approach. First, we calibrate the micro-foundation of the model—specifically the user's attention dynamics and utility parameters—using the *KuaiRand* dataset (Gao et al., 2022), a large-scale unbiased dataset collected from Kuaishou. Second, to tackle the computational intractability of the high-dimensional coupled Master Equation (Eq. 26), we propose a novel *Spectral Soft Actor-Critic (Spectral-SAC)* algorithm. This physics-informed Deep Reinforcement Learning (DRL) solver enables us to capture the complex non-local feedback loops of the traffic recycling mechanism efficiently.

4.1 Data and Structural Estimation

4.1.1 Data Source and Identification Strategy

To structurally identify the micro-foundation parameters of our model—specifically the natural attention decay rate μ and the intrinsic volatility σ—we utilize the *KuaiRand Dataset* (Gao et al., 2022). This is a large-scale sequential recommendation dataset collected from the video-sharing mobile App, Kuaishou.

The Identification Challenge: Endogeneity. A critical challenge in calibrating user attention dynamics from standard production logs is the presence of *endogeneity bias* (or confounding bias) (J. Chen et al., 2023). In a standard environment, the recommendation algorithm ($a(t, z)$) actively intervenes to match content with users. Consequently, the observed attention drift is a composite of intrinsic user patience and algorithmic matching quality:

$$\underbrace{\text{Observed Log-Drift}}_{\text{Data}} = \underbrace{\mu - \tfrac{1}{2}\sigma^2}_{\substack{\text{Unknown}\\\text{User Param}}} + \underbrace{a(t, z)}_{\substack{\text{Unobservable}\\\text{Algo Control}}} \tag{33}$$

where the log-drift $\mu - \frac{1}{2}\sigma^2$ is the Itô-corrected drift of $X_t = \ln Z_t$ under GBM (by Itô's formula: $\mathrm{d}X_t = (\mu + a - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma \mathrm{d}W_t$). The structural MLE in Section 4.1 recovers μ and σ jointly via the Inverse Gaussian moment conditions, correctly accounting for this correction. Without isolating the algorithmic intervention, the structural parameter μ remains unidentified, as a long dwell time could result from either high user patience or an exceptionally good recommendation.

Identification Strategy: The Randomized Experiment. To disentangle these effects, we exploit the unique *KuaiRand-Pure* subset (Gao et al., 2022). In this regime, videos were exposed to users via a purely random policy, effectively setting the algorithmic intervention to zero ($a(t, z) \approx 0$). This allows us to treat the observed viewing behaviors as a *Randomized Controlled Experiment*, revealing the users' baseline physiological and psychological reaction functions (the "Organic Baseline") without the confounding influence of the recommender policy (Saito, Yaginuma, Nishino, Sakata, & Nakata, 2020). This identification strategy provides the ground truth for the "Healthy" parameters used in Section 3.4.

4.1.2 Variable Construction and Processing

We focus on the granular interaction logs where the user's viewing duration is recorded in milliseconds (Gao et al., 2022). We define the mapping between the theoretical stopping time τ (from our optimal stopping model) and the observed data. Let τ_{obs} denote the observed viewing time and D denote the finite duration of the video. The relationship is given by:

$$\tau_{obs} = \min(\tau, D). \tag{34}$$

Handling Right-Censoring. This structure implies that the data is *Right-Censored*. An observation where a user watches until the end ($\tau_{obs} = D$) provides only partial information: we know the user's latent patience τ satisfies $\tau \geq D$, but the exact stopping threshold was not triggered. Ignoring this censoring mechanism—i.e., treating completion as an exact measure of interest—would lead to a systematic **underestimation** of the natural decay parameter μ (overestimating drift).

Data Filtering and Sample Selection. Data processing was performed on the *KuaiRand-1K* subset (Gao et al., 2022). We applied rigorous filters to ensure the structural validity of the calibration:

1. **Mechanical Noise Removal:** We excluded interactions shorter than 0.5s. This threshold corresponds to the physiological limit of the human motor system reaction time, filtering out accidental touches devoid of cognitive processing.

2. **Content Validity:** We removed videos shorter than 3 seconds[2], ensuring a meaningful consumption horizon.

3. **User Activity:** We excluded inactive users with fewer than 5 interactions to ensure sufficient data points for stable parameter identification.

The final sample used for structural calibration consists of 40,922 interaction events from 992 unique users under the random exposure policy. Table 1 summarizes the descriptive statistics.

4.1.3 Structural Estimation via Maximum Likelihood

Under the Geometric Brownian Motion assumption for the attention state Z_t, the logarithm of attention, $X_t = \ln(Z_t)$, follows an Arithmetic Brownian Motion. Consequently, the First Passage Time (FPT) to the swiping threshold b^* follows an *Inverse Gaussian (IG)* distribution (also known as the Wald distribution) (Folks & Chhikara, 1978; Peskir & Shiryaev, 2006).

The probability density function (PDF) for the random stopping time τ is given by:

$$f(t; \eta, \lambda) = \sqrt{\frac{\lambda}{2\pi t^3}} \exp\left(-\frac{\lambda(t-\eta)^2}{2\eta^2 t}\right), \tag{35}$$

[2] The 3-second threshold aligns with the platform's definition of "short_time_play" and ensures the content has sufficient duration to convey information.

Table 1: Descriptive Statistics of the Processed Sample (KuaiRand-Pure)

Variable	**N**	**Mean**	**Std. Dev.**	**Median**	**Max**
Viewing Time (τ_{obs}, s)	40,922	6.97	18.14	2.15	508.1
Video Duration (D, s)	40,922	108.46	104.81	80.78	1177.7
Structure Metric			**Value**		
Unique Users			992		
Censoring Rate (Completes)			3.33%		

Note: The sample excludes mechanical noise ($< 0.5s$) and inactive users. Viewing time is treated as right-censored if $\tau_{obs} = D$. The low censoring rate (3.33%) confirms that "swiping" is the dominant mode of termination.

where $\eta > 0$ represents the mean time to swiping (mean FPT) and $\lambda > 0$ is the shape parameter characterizing the variance.

Handling Censored Data. To account for the right-censoring structure identified in the previous step, we construct the log-likelihood function by combining the probability density for observed switches (uncensored, $\delta_i = 1$) and the survival function $S(t) = 1 - \int_0^t f(s)\mathrm{d}s$ for videos that ended before a swipe occurred (censored, $\delta_i = 0$):

$$\mathcal{L}(\eta, \lambda) = \sum_{i=1}^{N} \Big[\delta_i \ln f(t_i; \eta, \lambda) + (1 - \delta_i) \ln S(D_i; \eta, \lambda) \Big]. \tag{36}$$

We maximize $\mathcal{L}(\eta, \lambda)$ using the L-BFGS-B optimization algorithm to ensure positivity constraints on parameters (Byrd, Lu, Nocedal, & Zhu, 1995).

Recovering Micro-Parameters. The estimated distributional parameters $(\hat{\eta}, \hat{\lambda})$ are not our final goal; we use them to recover the *structural parameters* (μ, σ) of the user's attention dynamics. Let $x_0 = \ln(z_0/b^*)$ denote the normalized initial distance to the threshold. Based on the properties of the FPT for Brownian motion, the mapping is derived as:

$$\begin{aligned} \hat{\sigma} &= \frac{x_0}{\sqrt{\hat{\lambda}}}, \\ \hat{\mu} &= \frac{1}{2}\hat{\sigma}^2 - \frac{x_0}{\hat{\eta}}. \end{aligned} \tag{37}$$

This procedure allows us to identify the natural decay rate $\hat{\mu}$ and volatility $\hat{\sigma}$ that govern the *Organic Baseline* of user behavior, stripped of algorithmic manipulation.

4.1.4 Calibration Results and Interpretation

The structural estimation results are presented in Table 2. The calibrated parameters establish the "*Physiological Baseline*" of the user, providing the empirical foundation for the

Organic Equilibrium discussed in Section 3.4.

We highlight three key empirical findings regarding the human attention mechanism:

1. **The Inevitability of Boredom ($\mu < 0$):** We identify a significant negative drift $\mu = -0.0235$. This empirically validates the model's core assumption: without high-intensity stimuli, human attention naturally decays. This acts as the physiological gravity that algorithms must fight against.

2. **The Gambling Structure (σ):** The estimated volatility $\sigma = 0.3000$ is substantial relative to the drift. This reveals that the short-video ecosystem functions psychologically like a *Slot Machine* (Variable Ratio Schedule). The high variance implies that while the average video may be boring, the "fat tail" of occasional high-reward content keeps users engaged via the uncertainty of the next swipe.

3. **The Cost of Swiping (ϵ):** The macroscopic churn rate is identified as $\epsilon = 1.25\%$. This is estimated from the full KuaiRand-1K interaction logs (merging both Random and Standard exposure records) by defining a churn event as any inter-swipe gap exceeding 30 minutes within a user session, together with an implicit exit at the end of the observation window for each user. Formally, $\epsilon =$ (session breaks + unique users)/total swipes. This implies that every swipe carries a non-trivial risk of user exit. In our Stackelberg game, this ϵ represents the *structural constraint* that prevents the platform from infinite acceleration—if the algorithm pushes the swipe velocity too high, the cumulative churn will drain the user pool (the "Recycling Stagnation" effect).

These calibrated values provide the "physics engine" for our subsequent numerical simulations, ensuring that our policy analysis is grounded in the empirical reality of the KuaiRand dataset.

Table 2: Structural Parameters Calibrated from KuaiRand-Pure

Parameter	**Symbol**	**Value**	**Economic Interpretation**
Natural Drift	μ	-0.0235	Users exhibit a systematic decay in attention (boredom) of approx. 2.35% per second.
Volatility	σ	0.3000	High content heterogeneity creates significant uncertainty ("gambling" effect).
Churn Rate	ϵ	0.0125	The probability of exiting the App after a single swipe is 1.25%.
Init. Distance	x_0	0.50	Normalized initial log-distance $x_0 = \ln(z_0/b^*)$; fixed to 0.50 to satisfy the natural decay constraint ($\mu < 0$).

Note: Parameters are estimated using Maximum Likelihood Estimation (MLE) on the Inverse Gaussian distribution. x_0 is set to 0.5 to satisfy the natural decay constraint ($\mu < 0$).

Figure 3 visualizes the goodness-of-fit of our structural estimation. As demonstrated in the figure, our theoretical model successfully captures the two defining morphological features of short-video consumption:

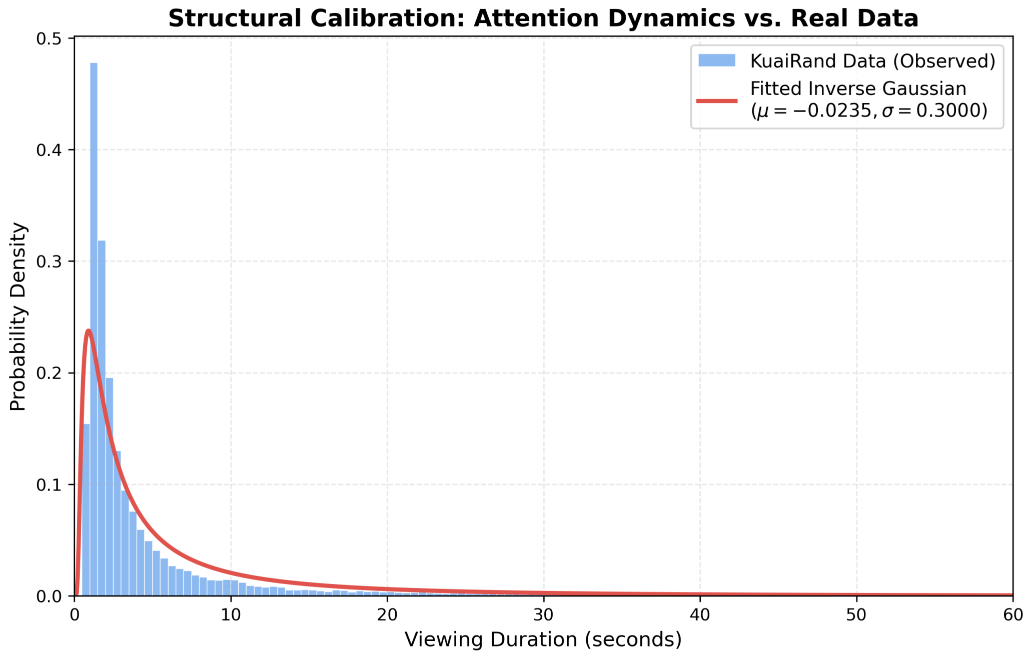

Figure 3: Structural Calibration Model Fit. The blue histogram represents the empirical probability density of user viewing durations observed in the KuaiRand-Pure dataset. The red solid line depicts the theoretical Inverse Gaussian (IG) distribution implied by our optimal stopping model, parametrized by the calibrated values $\hat{\mu} = -0.0235$ and $\hat{\sigma} = 0.3000$. The close alignment between the theoretical curve and empirical data validates the Geometric Brownian Motion assumption for attention dynamics.

Note: The comparison shows a strong fit in both the rapid dropout phase (left tail) and the heavy-tailed retention phase (right tail), confirming the model's ability to capture the dual nature of impatience and engagement.

1. **Rapid Physiological Filtering (Left Tail):** The model accurately reproduces the high probability mass concentrated in the initial few seconds. This reflects the "short attention span" characteristic, where users rapidly filter out content that fails to meet the minimum stimulation threshold within the first cognitive cycle ($< 2s$).

2. **Heavy-Tailed Engagement (Right Tail):** Crucially, the fitted curve aligns closely with the long right tail of the empirical distribution. This confirms that the calibrated volatility parameter ($\sigma = 0.3000$) effectively captures the "gambling-like" uncertainty mechanism: a small fraction of high-match videos can sustain user attention for extended periods despite the natural decay drift ($\mu = -0.0235$).

The strong concordance between the theoretical prediction and the natural experiment data provides robust empirical support for modeling user attention as a stochastic process with an absorbing boundary.

4.2 Solution Methodology: Spectral Deep Reinforcement Learning

Solving the Mean-Field Stackelberg Game described in Section 3 poses significant computational challenges. The system is characterized by a high-dimensional state space (the discretized density $m(t, \cdot)$) and, crucially, a *non-local coupling term* (traffic recycling) that introduces strong feedback loops. Standard numerical methods, such as finite difference schemes with adjoint-based optimization, suffer from the *curse of dimensionality* and often fail to converge due to the instability caused by the free boundary (Han, Jentzen, & E, 2018; Peskir & Shiryaev, 2006).

To overcome these hurdles, we develop a novel hybrid solver: *Spectral Soft Actor-Critic (Spectral-SAC)*. This approach integrates the structural regularization of spectral methods with the exploration capabilities of maximum entropy reinforcement learning.

Dimensionality Reduction via Spectral Embedding. A key innovation in our approach is the parameterization of the platform's infinite-dimensional control policy $a(t, \cdot) \in L^2(\Omega)$. Direct output for every spatial grid point (action dim $N_z \approx 400$) would lead to inefficient learning and jagged, discontinuous policies. Instead, we project the retention policy onto the subspace spanned by the first K Chebyshev polynomials $T_k(\cdot)$:

$$a(t, z) \approx \sum_{k=0}^{K-1} w_k(t) T_k(\tilde{z}). \tag{38}$$

This spectral technique (detailed in Online Appendix D) reduces the effective action dimension to $K + 4 = 14$ (the $K = 10$ spectral weights plus a global drift scalar, acquisition rate, matching investment, and content diversity control). Crucially, it acts as a *low-pass filter*, enforcing spatial smoothness on the algorithmic intervention, which aligns with the continuity requirements of practical recommendation systems.

Physics-Informed "Solver-in-the-Loop". Unlike black-box RL approaches that rely on purely data-driven transitions, our environment is *Physics-Informed*. At each interaction step, the reward calculation involves explicitly solving:

1. The user's Optimal Stopping problem (HJB QVI) to determine the swiping boundary $b^*(t)$.
2. The population's mass conservation (Non-Local Fokker-Planck equation) to compute the exact next state density.

This "Solver-in-the-Loop" architecture ensures that the learned policy strictly satisfies the user's incentive compatibility and the system's flux consistency constraints at all times, minimizing the gap between the RL approximation and the true MFG solution.

Theoretical Connection to the Master Equation. The Critic network $Q_\theta(s, a)$ in our algorithm serves as a function approximator for the value function of the Master Equation defined on the Wasserstein space of probability measures (Carmona, Delarue, et al., 2018). By minimizing the Bellman error using the Soft Actor-Critic (SAC) objective, the algorithm implicitly recovers the solution to the Mean-Field Game. The entropy regularization in SAC is particularly valuable here, promoting adequate exploration of the high-dimensional state space and improving the robustness of the solution against numerical instability.

For the detailed MDP formulation, network architecture, hyperparameter settings, and the complete pseudo-code (Algorithm 2), we refer readers to Online Appendix D.

5 Main Results: The Algorithmic Liquidity Trap

In this section, we present the numerical equilibrium results derived from the calibrated model. We contrast the equilibrium outcomes under two distinct monetization regimes: a *Retention-Oriented* strategy (maximizing time spent) and an *Ad-Centric* strategy (maximizing swipe flux).

Our analysis reveals a critical phase transition: when the incentive for ad impressions (λ_2) surpasses a critical threshold relative to retention (λ_1), the ecosystem collapses into a pathological state we term the *"Algorithmic Liquidity Trap."* In this regime, the platform creates an illusion of high engagement (extreme swipe velocity) that is, in reality, devoid of meaningful attention (Phantom Liquidity).

5.1 Experimental Setup and Parameter Specification

Physiological Baseline (Fixed). To ensure empirical validity, we fix the user behavioral parameters to the values structurally estimated from the KuaiRand dataset (Table 2). These represent the immutable *"Physiological Constraints"* of the user population:

- Natural Decay: $\mu = -0.0235$ (The force of boredom).
- Volatility: $\sigma = 0.3000$ (The gambling uncertainty).

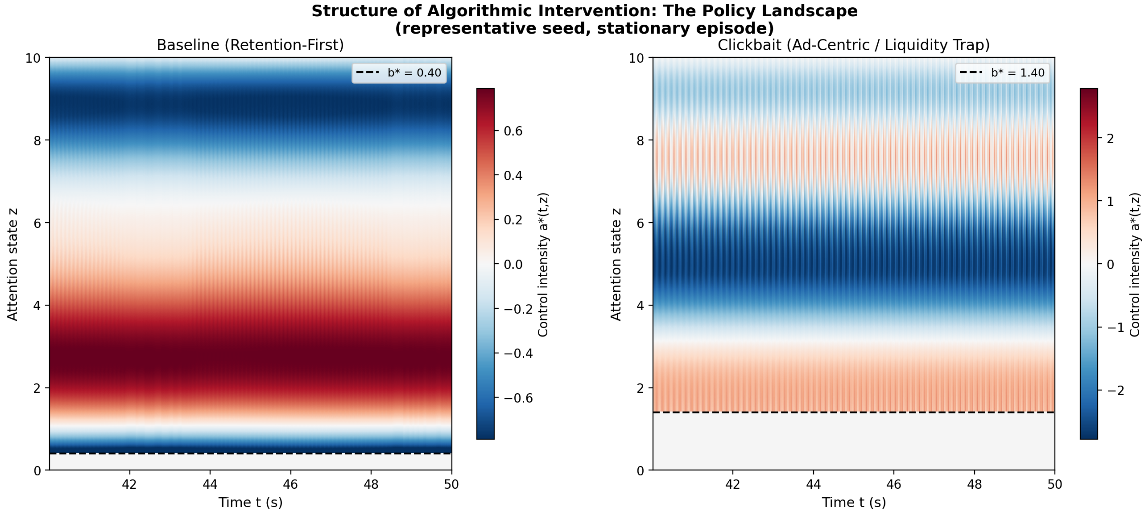

Figure 4: **Comparative Policy Landscape: Baseline vs. Algorithmic Liquidity Trap.** Both panels display the stationary-phase control intensity $a^*(t,z)$ (red = positive retention boost; blue = negative suppression) derived from the representative converged seed of each experiment. The dashed line marks the steady-state optimal stopping threshold b^*. **Left Panel (Baseline, λ_1=20, λ_2=0.1):** Under the retention-first objective, the policy applies moderate positive reinforcement across mid-range attention states ($z \approx$ 1–3) and mild suppression at high z, sustaining a low $b^* \approx 0.40$ well below the organic benchmark. Users are gently "locked in" to a deep-engagement regime. **Right Panel (Clickbait, λ_1=5, λ_2=5):** Under equal incentive weighting, a structurally distinct two-layer strategy emerges: a *Safety Net* layer of positive reinforcement just above b^* prevents total churn, while a *Planned Obsolescence* layer of strong negative control accelerates boredom at higher z, driving users back to the boundary in rapid succession. This "Trap-and-Release" cycle generates the extreme flux $\Phi^* \approx 236$ swipes/s and dwell time $\tau \approx 0.11$s that define the Algorithmic Liquidity Trap.

- Churn Rate: $\epsilon = 0.0125$ (The cost of recycling).

Platform Incentives (Variable). The core of our experiment lies in varying the platform's objective function (Eq. 21). We normalize the user's cognitive switching cost to $C = 1$. We define the *Monetization Ratio* $\mathcal{R}$ as the relative weight of impression revenue to retention value:

$$\mathcal{R} \equiv \frac{\lambda_2}{\lambda_1}. \tag{39}$$

Note on Scaling: Since the optimal control policy depends on the relative weights in the objective functional, our results are robust to the absolute scaling of λ_1 and λ_2. The phenomenon of the Liquidity Trap is driven by the high Monetization Ratio $\mathcal{R}$, independent of the total magnitude of the calculated profit.

We contrast two polar cases: the *Baseline* (λ_1=20, λ_2=0.1, so $\mathcal{R}$=0.005, retention-dominant) and the *Clickbait* regime (λ_1=5, λ_2=5, so $\mathcal{R}$=1, equal incentive weighting). Each configuration is trained across ten independent random seeds to assess equilibrium robustness; we report statistics over converged seeds only.

Detailed hardware specifications, hyperparameter settings for the SAC agent (e.g., learning rates, replay buffer size), and the spectral regularization parameter ($c_{\text{smooth}} = 20.0$) are provided in Online Appendix D.

Benchmarking the Solver. To validate the capability of our Spectral-SAC solver in handling the *complex feedback dynamics* of the ecosystem, we benchmarked it against a standard gradient-based optimization method (SLSQP). The comparison, detailed in Online Appendix **??**, confirms that traditional solvers are often limited to inferior linear policies. In contrast, the *exploratory nature* of our physics-informed DRL agent enables it to discover the globally superior non-linear control structures required to manipulate the equilibrium.

5.2 Baseline Equilibrium Analysis: The Healthy Retention Regime

We first examine the baseline scenario where the platform prioritizes user retention over immediate traffic flux (λ_1=20, λ_2=0.1, $\mathcal{R}$=0.005). The Spectral-SAC agent was trained for 500,000 steps across ten random seeds; 5 out of 10 seeds converged to a stationary equilibrium, and all quantitative results below are reported over these converged seeds. The simulation results, visualized in Figure 5, demonstrate that under this "User-Centric" policy the ecosystem successfully converges to a stable Mean-Field Equilibrium (MFE).

Sustainable Carrying Capacity (N^*). As shown in Figure 5 (A), the total active population $N(t)$ follows a logistic growth trajectory, stabilizing at $N^* = 73.0 \pm 6.4$ (mean $\pm$ 1 s.d. across converged seeds). Simultaneously, the traffic flux converges to a moderate stationary rate $\Phi^* = 2.5 \pm 1.7$ swipes/s. Via Little's Law ($\tau = N^*/\Phi^*$), this implies a median dwell time $\tau \approx 65.9\,$s—more than 30 times the IAB/MRC viewability threshold of 2 seconds. This stable, low-velocity flux serves as the "healthy heartbeat" of the ecosystem: user acquisition is organically balanced by endogenous churn without overheating attention.

The Lock-in Effect. A critical finding, depicted in Figure 5 (D), is the trajectory of the optimal stopping threshold b^*. Under the retention-oriented policy, the SAC agent learns to deploy a sustained positive retention field $a^*(t, z) > 0$ (Figure 5 (C)) that *lowers* the user's effective swiping threshold relative to the organic benchmark $b^*_\infty = 0.65$. The converged equilibrium yields $b^* = 0.514 \pm 0.077$, a statistically significant depression of the swiping threshold. We term this the *"Lock-in Effect"*: enhanced immediate gratification dominates the option value of switching, reducing b^* and extending the average dwell time. *Policy Implication:* The platform achieves ecosystem health not by raising content expectations, but by reducing users' perceived cost of staying. This sustains a large, engaged population and generates sufficient organic swipes to monetize effectively without artificially shortening attention spans—establishing the *Healthy Baseline* against which we measure the pathological effects of algorithmic greed.

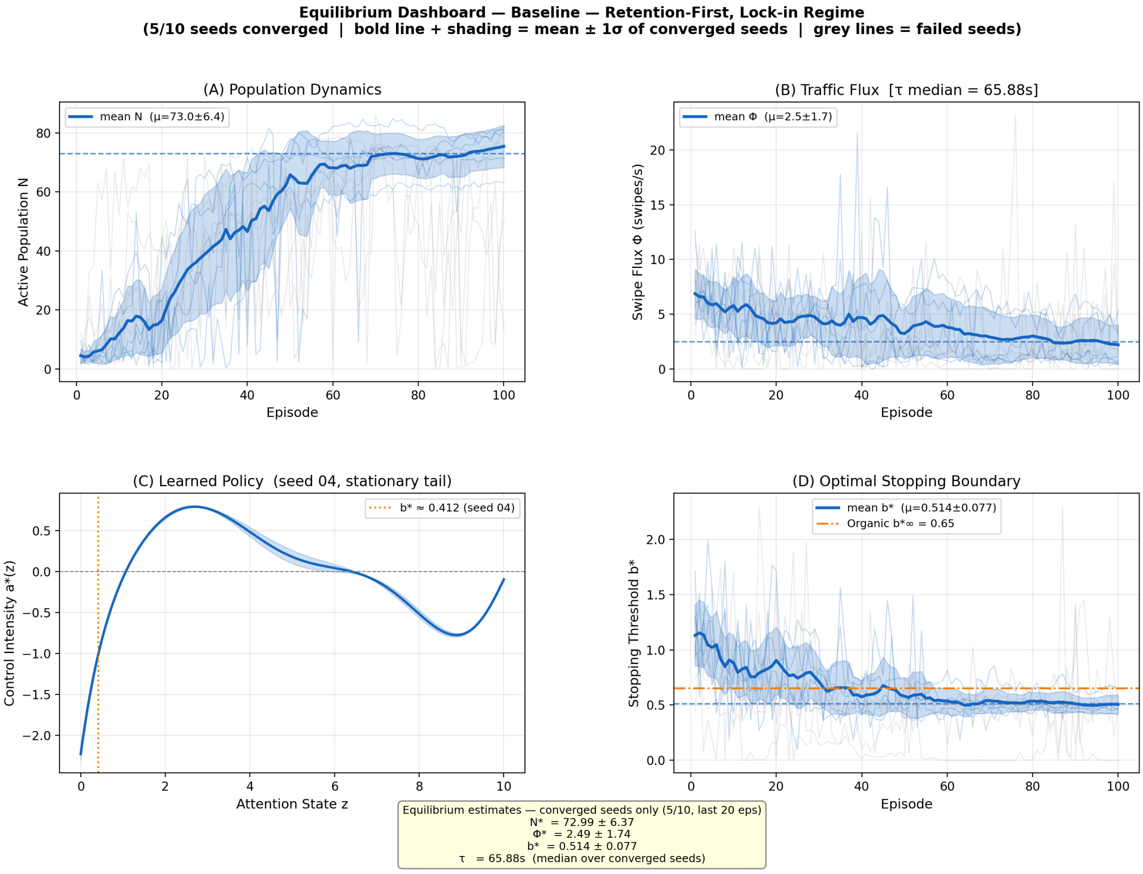

Figure 5: **The Healthy Baseline Equilibrium (λ_1=20, λ_2=0.1; 5/10 seeds converged).** Bold line and shaded band show the cross-seed mean $\pm 1\sigma$ of converged seeds; thin colored lines show individual converged trajectories; grey lines show non-converged seeds. **(A) Population Dynamics:** $N(t)$ converges to a sustainable carrying capacity $N^* = 73.0 \pm 6.4$. **(B) Traffic Flux:** $\Phi^* = 2.5 \pm 1.7$ swipes/s; median dwell time $\tau \approx 65.9$ s, far above the IAB/MRC 2-second viewability threshold. **(C) Learned Policy $a^*(z)$:** The optimal retention control exhibits a positive peak at intermediate z and mild suppression at high z, consistent with smooth spectral regularization. **(D) Lock-in Effect:** The optimal stopping threshold $b^* = 0.514 \pm 0.077$ (blue) is depressed below the organic benchmark $b^*_\infty = 0.65$ (orange dashed), confirming that persistent positive reinforcement reduces users' perceived option value of switching.

5.3 Comparative Statics: The Algorithmic Liquidity Trap

To investigate the dynamics of an Ad-Centric monetization strategy, we pivot the platform's objective from the "Editor" logic to the "Dealer" logic: we set $\lambda_1{=}5$, $\lambda_2{=}5$ ($\mathcal{R}{=}1$), placing equal weight on user retention and swipe-flux revenue. Seven out of ten seeds converge robustly; we report statistics over these converged seeds.

The results, visualized in Figure 6, reveal the emergence of a counter-intuitive and pathological regime we term the *"Algorithmic Liquidity Trap."*

Macroscopic Illusion: Phantom Liquidity. The steady-state population contracts to $N^* = 27.2 \pm 3.9$, less than 40% of the Baseline level, consistent with the "Leaky Bucket" mechanism. Yet this depleted user base generates a traffic flux of $\Phi^* = 235.8{\pm}20.5$ swipes/s—nearly **two orders of magnitude** above the Baseline (Figure 7). Via Little's Law, the implied average dwell time is $\tau \approx 0.11\,\text{s}$, a value **18 times below** the IAB/MRC viewability standard of 2 seconds.

Each user cycles through the content pool at extreme velocity, generating massive nominal traffic while spending negligible time on any single video. We define this state as *Phantom Liquidity*: the ecosystem is awash with traffic volume—potentially flattering headline DAU and impression metrics—but the actual attention depth per interaction has collapsed to a physiological floor well below any economically meaningful threshold.[3] Figure 8 quantifies the economic consequence: after applying the IAB/MRC viewability filter, Clickbait's billable flux collapses to near zero, and the Baseline strictly dominates on effective revenue for *every* ad-to-subscription revenue mix $\alpha \in [0, 1]$.

Micro-Strategy: The "Safety Net" and Planned Obsolescence. How does the algorithm achieve this high-velocity equilibrium? The learned control policy, as detailed in Figure 4 (Right Panel) and confirmed by Figure 6 (C), reveals a structurally distinct two-layer strategy:

- **Safety Net ($a^* > 0$ near b^*):** Immediately above the exit threshold the algorithm applies positive retention force to prevent total churn—keeping users inside the app.
- **Planned Obsolescence ($a^* < 0$ at higher z):** Once the user is safely retained, the algorithm actively *suppresses* attention quality, accelerating the natural boredom drift back toward the boundary and triggering another lucrative swipe event.

This "Trap-and-Release" cycle converts user *time* into *swipes* with surgical precision, exploiting the gambling-like volatility σ identified in Section 4.

[3] Such ultra-short exposures constitute *"Economic Zero"* under current industry standards. The Media Rating Council (MRC) classifies a mobile video impression as "Viewable" (and billable) only if it plays for at least 2 continuous seconds (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). Rapid dismissal also serves as a strong negative feedback signal (Active Rejection) in recommender systems, distinguishing it from merely "low-value" passive consumption.

The High-b^* Regime and Multiple Equilibria. A further distinguishing feature of the Clickbait equilibrium is the elevation of the optimal stopping threshold to $b^* = 2.554 \pm 0.706$—nearly **five times** the Baseline value and well above the organic benchmark $b^*_\infty = 0.65$ (Figure 6 (D)). The substantial cross-seed dispersion (± 0.706) is itself informative: it reflects the multiplicity of Nash equilibria that emerge under balanced incentive weighting ($\mathcal{R}$=1), consistent with the theoretical prediction that the transition region supports coexisting high-b^*/high-flux and low-b^*/low-flux fixed points. The economy has ceased to be a utility for content consumption and has evolved into a *"Slot Machine" mechanism*: profit is maximized not by extending the "Gaze" but by inducing rapid, repetitive cycles of anticipation and result.

Robustness Verification: Scale Invariance. A potential concern regarding Experiment A is whether the emergence of the "Liquidity Trap" is an artifact of the specific parameter scale ($\Sigma\lambda = 10$) rather than the monetization structure. To rule this out, we conducted a *Scale-Invariant Robustness Check* (Experiment C) where we doubled the incentive weights to $\lambda_1 = \lambda_2 = 10.0$. This normalizes the total weight budget to match the Baseline ($\Sigma\lambda = 20$) while maintaining the critical monetization ratio $\mathcal{R} = 1$.

The results, detailed in Online Appendix E, confirm the persistence of the trap. The system converged to a high-velocity equilibrium with flux Φ^* statistically consistent with the Ad-Centric regime and fundamentally distinct from the Baseline. This empirically confirms that the pathological state is driven by the *relative structure* of incentives ($\mathcal{R}$), independent of the absolute magnitude of the profit function.

6 Conclusion and Implications

This paper bridges the gap between the micro-foundations of user attention and the macro-dynamics of platform economics. By integrating an optimal stopping framework into a Mean-Field Stackelberg Game, and calibrating it with the KuaiRand dataset, we provide a structural lens to understand the "Scroll or Stop" dilemma.

6.1 Theoretical Contribution: The Anatomy of the Trap

Our central theoretical contribution is the identification of the *"Algorithmic Liquidity Trap."* We demonstrate that standard algorithmic objectives, when biased towards impression generation (the "Dealer" logic), do not merely optimize traffic; they fundamentally restructure the user's attention span.

- **Phantom Liquidity:** We show that high traffic metrics (Flux, impressions) can coexist with, and indeed mask, a collapse in engagement depth. The platform's ability to accelerate the *Traffic Recycling Loop* maintains aggregate throughput even as per-interaction attention depth collapses to a physiological floor—an illusion of prosperity driven by hyper-active churn rather than genuine satisfaction.

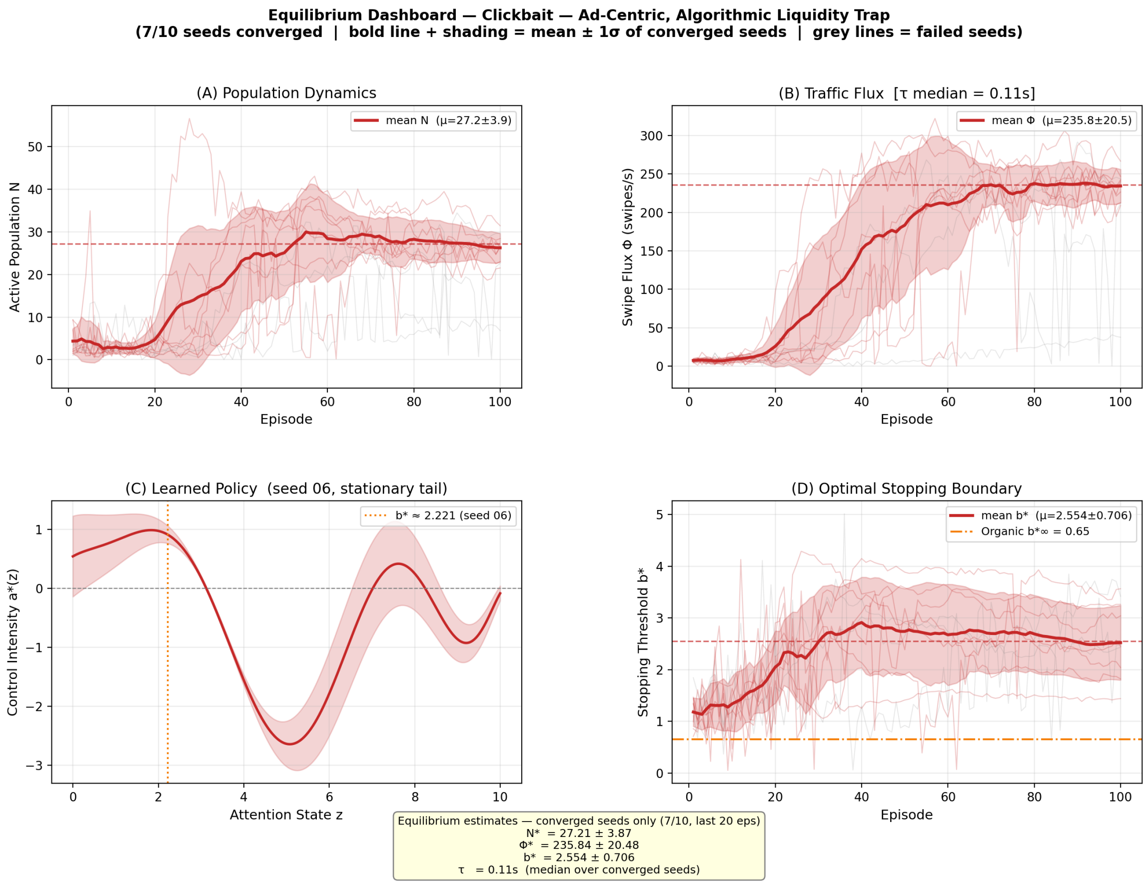

Figure 6: **The Algorithmic Liquidity Trap (λ_1=5, λ_2=5; 7/10 seeds converged).** Bold line and shaded band show the cross-seed mean $\pm 1\sigma$ of converged seeds; thin colored lines show individual converged trajectories; grey lines show non-converged seeds. **(A) Population Dynamics:** The user base contracts to $N^* = 27.2 \pm 3.9$, less than 40% of the Baseline level. **(B) Traffic Flux — Phantom Liquidity:** Despite the depleted population, $\Phi^* = 235.8 \pm 20.5$ swipes/s; median dwell time $\tau \approx 0.11\,\mathrm{s}$, 18× below the IAB/MRC 2-second viewability threshold. **(C) Learned Policy $a^*(z)$:** The two-layer Safety Net ($a^* > 0$ near b^*) + Planned Obsolescence ($a^* < 0$ at higher z) structure converts user time into swipes with surgical precision. **(D) High-b^* Regime:** The optimal stopping threshold $b^* = 2.554 \pm 0.706$ (red) is elevated far above both the organic benchmark $b^*_\infty = 0.65$ and the Baseline level, reflecting a shift to a high-impatience, high-flux equilibrium. The large cross-seed dispersion is consistent with the theoretical prediction of multiple equilibria near $\mathcal{R}$=1.

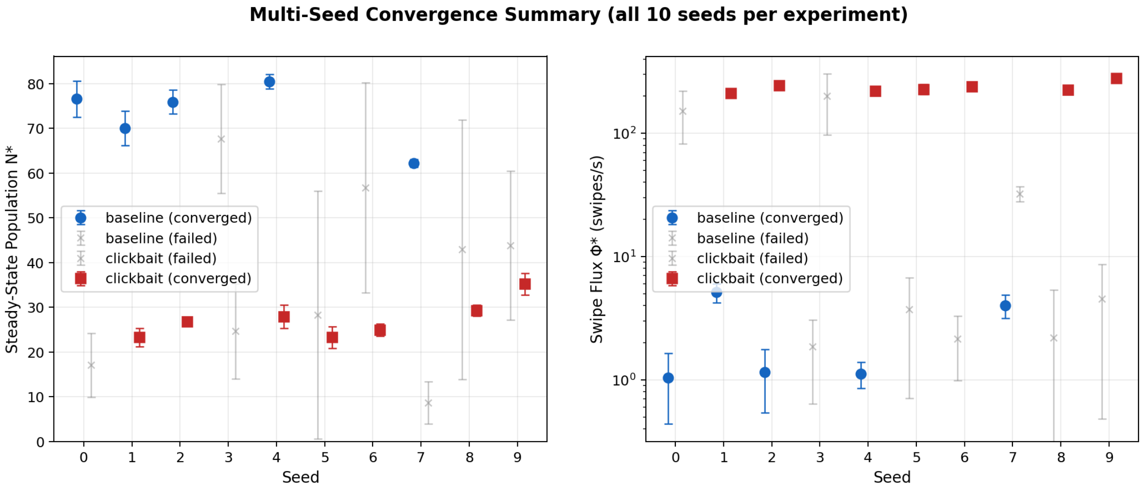

Figure 7: **Multi-Seed Convergence Summary.** Each point shows the steady-state estimate (mean $\pm$ 1 s.d. over the last 20 episodes) for one random seed. Filled symbols denote converged seeds; crosses denote non-converged seeds. **Left:** Steady-state population N^*. **Right:** Swipe flux Φ^* (log scale). The two regimes are non-overlapping on both dimensions, confirming the robustness of the qualitative contrast across seeds.

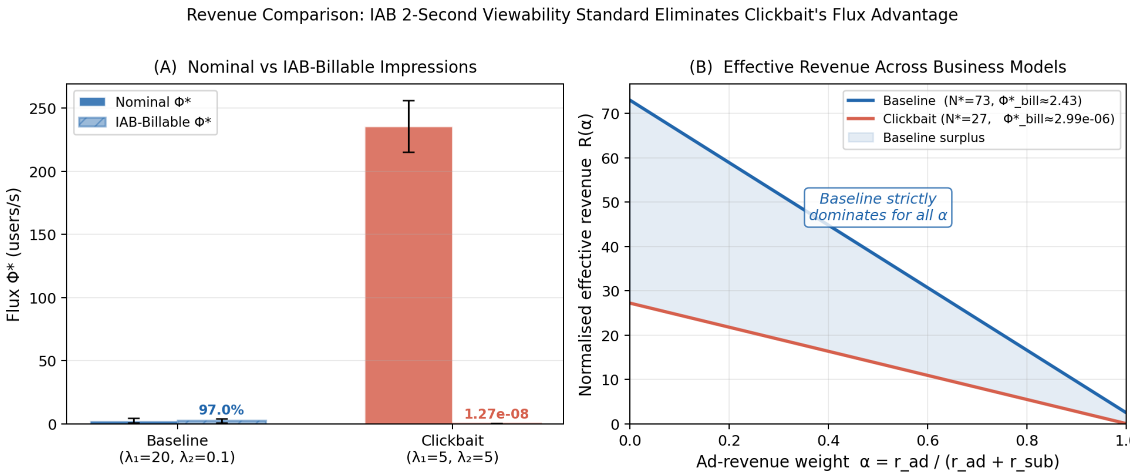

Figure 8: **Revenue Collapse Under the IAB/MRC Viewability Standard.** **(A)** Nominal swipe flux Φ^* versus IAB-billable flux $\Phi^*_{\text{bill}} = \Phi^* \cdot e^{-\tau_{\text{IAB}}/\tau}$ (with $\tau_{\text{IAB}} = 2\,\text{s}$). The Baseline retains 97.0% of its flux as billable impressions ($\tau \approx 65.9\,\text{s} \gg \tau_{\text{IAB}}$), while Clickbait's effective billable rate collapses to $\approx 10^{-8}$ ($\tau \approx 0.11\,\text{s} \ll \tau_{\text{IAB}}$). **(B)** Effective revenue $R(\alpha) = \alpha\,\Phi^*_{\text{bill}} + (1-\alpha)\,N^*$ for all ad-revenue weight $\alpha \in [0,1]$. The Baseline strictly dominates Clickbait for every possible ad-to-subscription revenue mix, resolving the apparent paradox of "phantom liquidity."

- **The Safety Net Mechanism:** We reveal how sophisticated reinforcement learning agents weaponize human physiology. By applying the "Safety Net" policy—relaxing control to induce decay and catching users only at the exit boundary—algorithms effectively commoditize user boredom, converting physiological impatience into commercial velocity.

6.2 Managerial Implications

For platform managers, our findings sound a cautionary note regarding the *KPI Alignment Problem*.

1. **The Metric Fallacy:** Optimizing for short-term flux (swipes/sec) yields immediate ad inventory gains (as seen in Experiment A) but pushes the ecosystem towards a high-volatility, low-trust equilibrium. This "Slot Machine" economy is fragile; while our model assumes a fixed physiological baseline, in reality, prolonged exposure to the "Safety Net" strategy may permanently erode user patience ($\mu \downarrow$), leading to eventual ecosystem collapse.

2. **Quality vs. Velocity:** We show that a "Retention-First" strategy (Baseline) is not antithetical to monetization. It generates *Organic Monetization* by maintaining a sustainable carrying capacity, validating that impression revenue and long dwell times are complements rather than substitutes when the Lock-in Effect operates. Managers must resist the temptation of "Algorithmic Overheating" to preserve the long-term health of the user pool.

6.3 Policy Implications

For regulators, this study suggests that the current scrutiny on "Addiction" (Total Time Spent) may be insufficient. The more insidious danger lies in *"Cognitive Acceleration."* Algorithms that maximize velocity do not necessarily keep users "hooked" for longer hours; instead, they condition users to process information at a frenetic pace, systematically stripping away the capacity for deep attention. Regulatory frameworks should consider monitoring *Attention Depth Metrics* (e.g., average dwell time per impression) alongside standard engagement figures to assess the cognitive health of digital platforms.

6.4 Limitations and Future Directions

While our framework offers significant insights, it abstracts away certain complexities of the real-world ecosystem.

1. **The Missing Supply Side (Creator Economy):** Our current model treats the content pool as a distribution $g(z; u)$ controlled effectively by the platform. In reality, content is generated by *Strategic Creators* who react to algorithmic incentives. If the platform falls into the "Algorithmic Liquidity Trap" (prioritizing flux), creators will rationally

respond by producing shorter, sensationalist content to maximize their own exposure. This *Supply-Side Adaptation* could create a secondary feedback loop, exacerbating the "Race to the Bottom" and making the ecosystem even harder to fix. Future work should model creators as a third player in this game.

2. **Competition and Network Effects:** We assume a single monopoly platform. A natural extension is to introduce *Platform Competition*, where rival algorithms compete for the same user attention budget. It remains an open question whether competition would force platforms to return to a "Quality-First" equilibrium or accelerate the acceleration towards even higher velocities. Additionally, incorporating social network effects (viral sharing) could offer new avenues for mechanism design beyond pure algorithmic matching.

3. **Heterogeneous User Types:** Our Mean-Field approach assumes a representative agent structure (with stochastic heterogeneity). Introducing distinct user types (e.g., "Passive Viewers" vs. "Active Searchers") could reveal distributional consequences, where the algorithm might exploit vulnerable sub-populations (e.g., minors with lower μ) while serving high-quality content to more selective users.

4. **Co-evolutionary Dynamics and Strategy Expansion:** Our Stackelberg framework models users as rational agents optimizing within a fixed physiological attention mechanism. In reality, the interaction may resemble a dynamic repeated game where users "learn" to counter algorithmic manipulation over time (e.g., developing "ad-blindness" or actively skipping content). However, we argue that the *Algorithmic Liquidity Trap* is robust because it exploits invariant physiological constraints (boredom decay μ and reward volatility σ) rather than transient behavioral loopholes. While users may evolve cognitive countermeasures, the platform's superior adaptation speed (Leader advantage) allows it to maintain the trap by shifting the locus of extraction. Future work could explicitly model this "Arms Race" using Evolutionary Game Theory.

5. **Internalizing Industry Viewability Standards into Platform Design:** Our analysis demonstrates ex-post, via the IAB/MRC-adjusted revenue function $R(\alpha) = \alpha \cdot \Phi^* e^{-\tau_{\text{IAB}}/\tau} + (1-\alpha) \cdot N^*$, that the Clickbait equilibrium is economically self-defeating once advertiser viewability standards are enforced. A natural and practically important extension is to embed τ_{IAB} *directly into the platform's training objective*—replacing raw flux Φ^* with billable flux $\Phi^*_{\text{bill}} = \Phi^* \cdot e^{-\tau_{\text{IAB}}/\tau}$ as the revenue signal. This "Compliance-Aware" platform objective transforms the regulator's standard into an endogenous constraint, and our framework predicts it would steer the learned policy away from the Liquidity Trap toward a healthier equilibrium closer to the Baseline. Training such a constrained DRL agent and characterizing its equilibrium properties constitutes a promising avenue for mechanism design on short-video platforms.

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A Mathematical Proofs

A.1 Proof of Proposition 2 (Threshold Structure)

The proof relies on the classical theory of optimal stopping for one-dimensional diffusions.

Let $\mathcal{L}^a$ denote the infinitesimal generator of the diffusion process Z_t. We consider the *waiting region* (continuation region) $\mathcal{C}_t = \{z \in \Omega \mid v(t,z) > K(t,u) - C\}$. Based on the results in Peskir and Shiryaev (2006) and Stokey (2008), the shape of the stopping region is determined by the monotonicity of the "local benefit of waiting".

For the CRRA utility function $u_{util}(z) = \frac{z^{1-\gamma}}{1-\gamma}$ with $\gamma > 1$ and the geometric Brownian motion dynamics, the net flow utility (holding cost vs. immediate reward) satisfies the single-crossing property. Specifically, the infinitesimal generator applied to the utility flow exhibits monotonicity in z: the marginal value of "waiting" (staying on the video) strictly increases with the attention state z.

Since the diffusion is non-degenerate ($\sigma > 0$), the Principle of Smooth Fit applies. The monotonicity ensures that the continuation region $\mathcal{C}_t$ is connected and takes the form of an upper set:

$$\mathcal{C}_t = (b^*(t), z_{\max}], \tag{40}$$

where the threshold $b^*(t)$ is the unique point satisfying the smooth-pasting condition. Consequently, the optimal stopping region is the complement $\mathcal{S}_t = [z_{\min}, b^*(t)]$. □

B Numerical Solution Details

B.1 A Fully Implicit Scheme with Mass Truncation

Given the implicit obstacle structure of the Quasi-Variational Inequality (QVI) and the non-local nature of the population dynamics, obtaining an analytic solution is mathematically infeasible. Therefore, this study constructs a numerical solution framework based on the *Fully Implicit Finite Difference Method (FDM)*. We adapt the iterative approach used in impulse control problems (Bensoussan & Lions, 1984) to handle the specific coupling between the user's stopping boundary and the traffic recycling mechanism, leveraging established schemes for Mean Field Games (Achdou, Camilli, & Capuzzo-Dolcetta, 2012; Achdou & Capuzzo-Dolcetta, 2010).

Approximation of the Infinite-Horizon Solution Although the theoretical model is formulated on an infinite time horizon $t \in [0, \infty)$, numerically resolving the system requires a finite temporal truncation. We approximate the infinite-horizon stationary equilibrium by simulating the coupled system over a sufficiently large finite time horizon $T_{\text{sim}} = 50$. This truncation is justified by two factors:

1. **Discounting Effect:** Given the discount rate ρ, the contribution of utility flows from $t > T_{\text{sim}}$ to the value function at $t = 0$ decays exponentially (proportional to $e^{-\rho T_{\text{sim}}}$), rendering the truncation error negligible.

2. **Convergence to Stationarity:** Our numerical experiments leverage the *Turnpike Property* of optimal control problems (Trélat & Zuazua, 2015). As shown in the results, the system dynamics (both the optimal boundary $b^*(t)$ and population density $f(t, \cdot)$) rapidly converge to a time-invariant state well before the terminal time. The time-stepping scheme thus serves as a fixed-point iteration to recover the stationary Mean-Field Equilibrium (MFE).

To eliminate the artifacts caused by the artificial terminal condition at T_{sim} (the terminal boundary layer), we report and analyze the solution over the interval $t \in [0, 0.8T_{\text{sim}}]$, where the system exhibits stable equilibrium behavior.

(1) Discretization Framework and Computational Domain In the mathematical model, the evolution of the density effectively takes place on the time-dependent (physical) domain $\Omega_t = [b^*(t), z_{\max}]$, since users who hit the endogenous swiping threshold $b^*(t)$ stop and are removed from the current video. Directly solving PDEs on a moving domain is computationally costly.

Therefore, we adopt a *Fixed (Fictitious) Domain Method*. We embed the problem into a fixed computational domain $\Omega = [z_{\min}, z_{\max}]$, discretized into a uniform grid $\mathcal{Z}_h = \{z_i \mid z_i = z_{\min} + i\Delta z\}_{i=0}^{N_z}$. The time interval is discretized into N_t time steps $\mathcal{T}_h = \{t_n \mid t_n = n\Delta t\}_{n=0}^{N_t}$. To ensure numerical stability given the stiff source terms (Dirac delta) and the recycling flux, a *fully implicit time integration scheme* is employed (LeVeque, 2007).

Boundary and Projection Implementation. We solve the Fokker–Planck equation on the entire fixed grid with *zero-flux (reflecting) boundary conditions* at $z_{\min}$ and $z_{\max}$ to avoid artificial mass leakage through the computational container. The interaction between the free boundary $b^*(t)$ and the fixed grid is handled by a projection ("cut") step: after each implicit update, we identify the region $z < b^*(t_n)$, integrate the mass in this region to compute the swipe flux, and then set the density to zero on $z < b^*(t_n)$. The removed mass is partially reinjected via $g(z; u)$ (traffic recycling) and partially lost via churn.

(2) Backward Solution of the User HJB Equation For a fixed platform policy $\{a(t, z), u(t)\}$, the user's value function $v(t, z)$ is solved via backward time stepping. For the convection term in the operator $\mathcal{L}^a v$, an upwind difference scheme is adopted to ensure the monotonicity of the scheme (M-matrix property), a critical condition for convergence in stochastic control problems (Kushner, 1990):

$$\partial_z v(z_i) \approx \begin{cases} \frac{v_{i+1} - v_i}{\Delta z} & \text{if } \mu + a(t_n, z_i) \geq 0 \\ \frac{v_i - v_{i-1}}{\Delta z} & \text{if } \mu + a(t_n, z_i) < 0 \end{cases}$$

The optimal stopping constraint is enforced via a projection operation at each time step, similar to the Projected Successive Over-Relaxation (PSOR) method for obstacle problems (Glowinski, 2013):

$$v_i^n = \max\{\tilde{v}_i^n, K(t_n, u^n) - C\}$$

where $K(t_n, u^n)$ is the endogenous outside option computed using the current matching control $u(t_n)$. The optimal stopping boundary $b^*(t_n)$ is identified as the critical state separating the continuation and stopping regions.

(3) Forward Solution of the Non-Local FPE Unlike standard decoupled frameworks, our population dynamics involve a *feedback loop* via the boundary flux. Once $b^*(t)$ is obtained, we solve the Non-Local Fokker-Planck Equation forward in time using a conservative *Fully Implicit Scheme*.

Flux Calculation via Mass Truncation: Instead of computing the gradient at the boundary, we calculate the swipe flux Φ_{swipe} by explicitly integrating the probability mass that falls into the stopping region $z < b^*(t_n)$ during the implicit step. This approach aligns with the conservative flux reconstruction methods proposed by Chang and Cooper (1970):

$$\Phi_{\text{swipe}}(t_n) = \frac{1}{\Delta t} \sum_{z_i < b^*(t_n)} f^*(t_n, z_i) \Delta z$$

This mass is physically removed from the system and a fraction $(1 - \epsilon)$ is recycled back via the source term $g(z; u)$. This ensures rigorous mass conservation and stability.

(4) Iterative Resolution of the QVI Fixed Point Since the obstacle $K(t, u)$ depends on the value function v itself, a fixed-point iteration is required (Bensoussan & Lions, 1984):

- **Initialization:** Guess an initial trajectory for $K^{(0)}(t)$.
- **Update Step:** In iteration m, solve HJB backward to get $v^{(m)}$, then update the coupling variable:
$$K_{\text{new}}(t) = \int_{z_{\min}}^{z_{\max}} v^{(m)}(t, z) g(z; u(t)) dz$$
- **Convergence:** Apply relaxation until $\|K^{(m+1)} - K^{(m)}\|_\infty < \text{Tol}$.

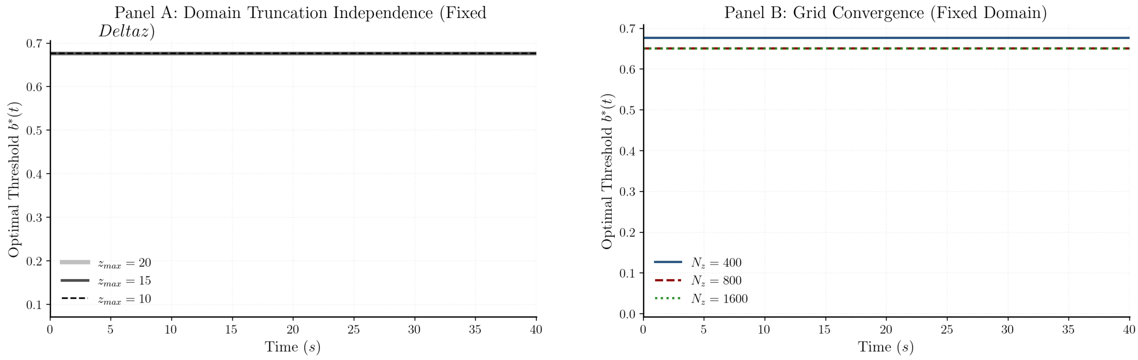
B.2 Numerical Robustness and Convergence

To ensure that our equilibrium results are not artifacts of the numerical discretization or the domain truncation, we performed rigorous sensitivity analyses, as illustrated in Figure 9.

Domain Independence (Panel A). We first verify the validity of the reflecting boundary condition at $z_{\max}$. We extended the computational domain from the baseline $z_{\max} = 10$ to $z_{\max} = 20$ while strictly maintaining the grid resolution Δz. As shown in Figure 9(a), the resulting optimal threshold curves $b^*(t)$ overlap closely. This confirms that the choice of $z_{\max} = 10$ is sufficiently large to eliminate boundary reflection artifacts.

Grid Convergence (Panel B). We explicitly test for discretization errors by refining the spatial grid $N_z \in \{400, 800, 1600\}$ while fixing $z_{\max} = 10$. Figure 9(b) demonstrates convergence of the solution. Notably, the solutions for $N_z = 800$ and $N_z = 1600$ are virtually indistinguishable, indicating that the solver reaches mesh independence at $N_z \approx 800$.

While the baseline resolution used in exploratory simulations ($N_z = 400$) exhibits a minor quantitative offset due to discretization viscosity, it preserves the qualitative structure and dynamic trends of the converged solution.



(a) Domain truncation independence.

(b) Grid convergence.

Figure 9: **Numerical robustness and convergence checks.** Panel (a) holds Δz fixed while expanding the computational domain ($z_{\max} = 10, 15, 20$) and shows that $b^*(t)$ is insensitive to domain truncation. Panel (b) fixes $z_{\max} = 10$ and refines the grid ($N_z = 400, 800, 1600$), demonstrating mesh convergence of the threshold dynamics.

B.3 Parameter Configuration and Pseudocode

The model parameters are calibrated to ensure a stable S-curve growth trajectory and a meaningful stationary equilibrium. Table 3 details the specific settings used in the simulation. Algorithm 1 presents the detailed numerical procedure.

Table 3: Model Parameters (Calibrated for Stability)

Symbol	Value	Description
User & Content Parameters		
γ	2.1	Relative risk aversion
ρ	0.5	Time discount rate (High impatience)
μ	-0.0235	Natural attention decay (Boredom)
σ	0.3	Content volatility
C	1.0	Cognitive switching cost
Ecosystem Parameters		
ϵ	0.0125	Churn Rate (1.25% exit probability)
A	5.0	Base Acquisition Rate
$N_{\max}$	100.0	Theoretical Market Capacity
μ_g	1.65	Content Quality Mean
Numerical Settings		
T	50.0	Simulation Time Horizon
Δt	0.01	Time Step Size
N_z	400	Grid Points ($\Delta z = 0.025$)

Note: Parameters are calibrated to ensure a stable S-curve growth trajectory.

Algorithm 1 Fully Implicit Solver for User-Population Equilibrium

procedure SOLVEEQUILIBRIUM(Policy $\{a, u\}$, Params $A, \epsilon, N_{\max}$)
  Input: Control $\{a, u\}$, Market Params. **Output:** v^*, b^*, f^*, Φ
  Phase 1: User Optimization (Backward HJB)
  Construct implicit operators; Initialize implicit system.
  for $n \leftarrow N_t - 1$ **downto** 0 **do**
    Solve linear system for $\tilde{v}$; $v^n \leftarrow \max(\tilde{v}, K(t_n) - C)$
    $b^*(t_n) \leftarrow \min\{z \mid v(t_n, z) > K(t_n) - C\}$ ▷ Identify threshold
    if Endogenous K **then** Update $K(t_n)$ via fixed-point iteration
    end if
  end for
  Phase 2: Population Dynamics (Forward FPE)
  Initialize $f(0, z)$ and $N(0)$; Construct **Fully Implicit Matrix** $\mathbf{M}_{FPE}$
  for $n \leftarrow 0$ **to** $N_t - 1$ **do**
    1. Acquisition: $\alpha \leftarrow \max(0, 1 - N^n/N_{\max})$; $S_{new} \leftarrow \alpha A \delta(z - z_p)$
    2. Implicit Solve: $\mathbf{M}_{FPE} \cdot \mathbf{f}^* = \mathbf{f}^n + \Delta t \cdot S_{new}$
    3. Flux & Cut: $M_{stop} \leftarrow \sum_{z<b^*} \mathbf{f}^* \Delta z$; $\Phi(t_n) \leftarrow M_{stop}/\Delta t$
      Set $\mathbf{f}^*(z) = 0$ for all $z < b^*(t_{n+1})$ ▷ Truncate distribution
    4. Recycling: $\mathbf{f}^{n+1} \leftarrow \mathbf{f}^* + (1 - \epsilon)\Phi(t_n) g(z) \Delta t$; Update $N(t_{n+1})$
  end for
  return $\{v, b, f, \Phi\}$
end procedure

C Derivation of the Master Equation on the Space of Measures

In this appendix, we derive the Hamilton-Jacobi-Bellman (HJB) equation for the platform's value function $V(N, m)$, utilizing the structure-scale decomposition introduced in Section 3.3.1.

C.1 Variational Derivatives and Chain Rule

Let $\mathcal{W}(f)$ be the value functional on the space of unnormalized measures. We decompose $f = N \cdot m$. To establish the Master Equation in terms of (N, m), we first relate the functional derivative $\frac{\delta \mathcal{W}}{\delta f}$ to the derivatives with respect to N and m.

Consider a perturbation $f \to f + \delta f$. The variations in N and m are:

$$\delta N = \int \delta f(z) dz, \quad \delta m(z) = \frac{\delta f(z)}{N} - \frac{f(z)}{N^2} \delta N.$$

The total variation of the value function is:

$$\begin{aligned} \delta \mathcal{W} &= \frac{\partial V}{\partial N} \delta N + \int \frac{\delta V}{\delta m}(z) \delta m(z) \, \mathrm{d}z \\ &= \int \left[\frac{\partial V}{\partial N} + \frac{1}{N} \left(\frac{\delta V}{\delta m}(z) \right. \right. \\ &\qquad \left. \left. - \int \frac{\delta V}{\delta m}(y) m(y) \, \mathrm{d}y \right) \right] \delta f(z) \, \mathrm{d}z. \end{aligned} \tag{41}$$

Thus, the shadow price of the unnormalized density is given by the *Decomposition Identity*:

$$\begin{aligned} \frac{\delta \mathcal{W}}{\delta f}(z) &= \frac{\partial V}{\partial N} + \frac{1}{N}\hat{\lambda}(z), \\ \text{where } \hat{\lambda}(z) &= \frac{\delta V}{\delta m}(z) - \mathbb{E}_m\left[\frac{\delta V}{\delta m}\right]. \end{aligned} \tag{42}$$

C.2 The Bellman Principle

The dynamic programming principle states:

$$\begin{aligned} rV(N,m) = \sup_{a,u,A} \Big\{ &\pi(N,m,a,u,A) \\ &+ \frac{\partial V}{\partial N}\frac{dN}{dt} + \left\langle \frac{\delta V}{\delta m}, \frac{\partial m}{\partial t} \right\rangle \Big\} \end{aligned} \tag{43}$$

Substituting the dynamics derived in the main text:

- **Scale Dynamics:** $\frac{dN}{dt} = A - \epsilon N \phi_{swipe}(m)$
- **Structural Dynamics:** $\frac{\partial m}{\partial t} = \mathcal{L}^*[a]m + \frac{A}{N}(\delta_{z_p} - m) + (1-\epsilon)\phi_{swipe}(m)g + \epsilon\, \phi_{swipe}(m)m$

C.3 Optimization of Controls

Before optimizing specific controls, we must explicitly handle the duality pairing involving the differential operator $\mathcal{L}^*[a]$.

Green's Identity and Boundary Terms: The term $\langle \frac{\delta V}{\delta m}, \mathcal{L}^*[a]m \rangle$ involves the Fokker-Planck operator. Using Green's identity (integration by parts twice) on the domain $\Omega = [b^*, z_{max}]$ with the absorbing boundary condition $m(b^*) = 0$:

$$\begin{aligned} \int_{b^*}^{z_{\max}} \frac{\delta V}{\delta m}\mathcal{L}^*[a]m\, dz = &\int_{b^*}^{z_{\max}} \mathcal{L}^a\left[\frac{\delta V}{\delta m}\right] m\, dz \\ &- \frac{\delta V}{\delta m}(b^*) \cdot \underbrace{\frac{1}{2}\sigma^2(b^*)^2\partial_z m(b^*)}_{\phi_{\text{swipe}}(m)} \end{aligned} \tag{44}$$

The boundary term $-\frac{\delta V}{\delta m}(b^*)\phi_{swipe}(m)$ arises because while the density m vanishes at the absorbing boundary, its gradient (flux) is non-zero. This term represents the *value destruction* of a user hitting the churn threshold. Note that we assume zero flux at the reflective boundary z_{max}.

We now maximize the Hamiltonian term by term:

1. Retention Control $a(z)$: The drift control $a(z)$ appears in the generator $\mathcal{L}^a[\frac{\delta V}{\delta m}] = (\mu + a(z))z\partial_z(\frac{\delta V}{\delta m}) + \dots$. Combining with the cost term $-\frac{\xi_1}{2}N\int a^2 m dz$:

$$\sup_a \int \left(-\frac{\xi_1}{2}Na^2 + az\partial_z\frac{\delta V}{\delta m}\right) m dz \implies a^*(z) = \frac{z}{\xi_1 N}\partial_z\frac{\delta V}{\delta m}$$

2. Acquisition Control A: Terms involving A:

$$A\left(\frac{\partial V}{\partial N}+\frac{1}{N}\frac{\delta V}{\delta m}(z_p)-\frac{1}{N}\int\frac{\delta V}{\delta m}mdz\right)-\Psi(A)$$

Using the identity (42), the coefficient of A is simply $\frac{\delta\mathcal{W}}{\delta f}(z_p)$. The maximum is $H_{acq}(\frac{\delta\mathcal{W}}{\delta f}(z_p))$.

3. Matching Control u: Terms involving u (via $g(z;u)$ and costs) and the flux ϕ_{swipe} (which collects the revenue, recycling, scale loss, *and boundary value loss* terms):

$$\begin{aligned}\text{Terms}=\phi_{\text{swipe}}\Bigg[&\underbrace{\lambda_2 N}_{\text{Ads}}+\underbrace{(1-\epsilon)\int\frac{\delta V}{\delta m}g(z;u)\,dz}_{\text{Recycling Gain}}\\&-\underbrace{(1-\epsilon)\int\frac{\delta V}{\delta m}m\,dz}_{\text{Mean Value Correction}}-\underbrace{\epsilon N\frac{\partial V}{\partial N}}_{\text{Churn Loss}}\\&-\underbrace{\frac{\delta V}{\delta m}(b^*)}_{\text{Boundary Loss}}\Bigg]-\frac{\xi_2}{2}u^2\end{aligned}\tag{45}$$

Let $\Delta V_{swipe}(u)$ denote the net marginal value of a swipe (the term in brackets). The optimal matching u^* satisfies:

$$\xi_2 u^*=\phi_{swipe}(1-\epsilon)\int\frac{\delta V}{\delta m}\frac{\partial g}{\partial u}(z;u^*)dz$$

C.4 The Final Equation

Collecting all terms yields the explicit Master Equation presented in Eq. (26). The inclusion of the boundary term $-\frac{\delta V}{\delta m}(b^*)\phi_{swipe}$ ensures the mathematical consistency of the Mean-Field Game on a domain with absorbing boundaries.

D Numerical Solution Details

D.1 MDP Formulation and Spectral Action Embedding

We reformulated the Mean Field Control (MFC) problem as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S},\mathcal{A},\mathcal{P},\mathcal{R},\gamma)$ as follows:

State Space $\mathcal{S}$. The state at time t is represented by the macroscopic configuration of the system, comprising the total population and the discretized density distribution of user attention:

$$\mathbf{s}_t=[N(t),m(t,z_1),m(t,z_2),\ldots,m(t,z_{N_z})]^\top\in\mathbb{R}^{N_z+1}\tag{46}$$

where $\{z_i\}_{i=1}^{N_z}$ are the spatial grid points. This high-dimensional state input allows the neural network to capture the fine-grained structure of the population distribution (Carmona & Laurière, 2021).

Spectral Action Space $\mathcal{A}$. A core challenge is that the action $a(t, \cdot)$ is a function in an infinite-dimensional space $L^2(\Omega)$. To mitigate the *curse of dimensionality* (Bellman, 1966), we employ a spectral parameterization method. We project the control function onto a subspace spanned by the first K Chebyshev polynomials $T_k(\cdot)$ (Boyd, 2001):

$$a(t, z) \approx \sum_{k=0}^{K-1} w_k(t) T_k(\tilde{z}), \quad \tilde{z} = \frac{2z - z_{max}}{z_{max}} \in [-1, 1] \tag{47}$$

Consequently, the action vector output by the agent is finite-dimensional:

$$\mathbf{u}_t = [u(t), A(t), w_0(t), \dots, w_{K-1}(t)]^\top \in \mathbb{R}^{K+2} \tag{48}$$

This spectral embedding reduces the action dimension from $N_z \approx 400$ to $K+2 \approx 12$, while naturally enforcing the spatial smoothness of the algorithmic intervention $a(t, z)$, which is crucial for the stability of the HJB solver.

Reward Function $\mathcal{R}$. The immediate reward reflects the platform's instantaneous profit flow:

$$r_t = N(t) \left(\lambda_1 \bar{z}_{ret} + \lambda_2 \Phi_{swipe}\right) - \left(\frac{\xi_1}{2}\|a\|_2^2 + \frac{\xi_2}{2}u^2 + \Psi(A)\right) \tag{49}$$

where $\bar{z}_{ret}$ is the average retention utility and Φ_{swipe} is the flux of swiping users.

D.2 Algorithm: Spectral Soft Actor-Critic

We adopt the Soft Actor-Critic (SAC) algorithm (Haarnoja, Zhou, Abbeel, & Levine, 2018), an off-policy maximum entropy DRL method. The interaction between the Agent (Platform) and the Environment (User Population) is governed by a physics-based engine that solves the coupled HJB-FPK system at each time step. The complete solution procedure is detailed in Algorithm 2.

D.3 Computational Infrastructure

To ensure the reproducibility of our numerical results, we detail the hardware and software environment used for the simulations and reinforcement learning training processes.
Hardware Specifications. All computations were executed on a high-performance computing server equipped with:

- **GPU:** NVIDIA Tesla P100 (16GB VRAM), ensuring efficient parallel computation for the neural network updates.
- **CPU:** Intel Xeon Platinum 8260 (2.30GHz, 12 vCPUs).
- **Memory:** 60GB RAM to handle the storage of the Experience Replay Buffer (size 10^6 transitions).

Algorithm 2 Spectral SAC with Physics-Informed Environment

Input: Basis $\{T_k\}_{k=0}^{K-1}$; Params $N_{\max}, \epsilon$; Rates α, β **Output:** Policy π_ϕ^*, Value Q_θ^*
Initialize Actor π_ϕ, Critic Q_θ, and Replay Buffer $\mathcal{D}$
for episode $m = 1$ **to** M **do**
  Initialize state $\mathbf{s}_0 = [N_0, \mathbf{f}_0]$
  for step $t = 0$ **to** $T_{horizon}$ **do**

    Phase 1: Spectral Action (Leader). Sample $\mathbf{u}_t \sim \pi_\phi(\cdot|\mathbf{s}_t)$ and reconstruct field:
      $a(z) \leftarrow \sum_{k=0}^{K-1} w_k T_k(\tilde{z})$ where $\mathbf{u}_t = [u, A, \{w_k\}]$ and $\tilde{z} \in [-1, 1]$
    Phase 2: User Equilibrium (HJB). Solve QVI via fixed-point iteration:
      Find $v(z)$ s.t. $\min(\mathcal{L}v - \rho v + U, v - (K - C)) = 0$
      Identify threshold: $b_t^* \leftarrow \min\{z \mid v(z) > K(u) - C\}$

    Phase 3: Population Dynamics (Implicit FPE).
      1. Acquisition: $S_{new} \leftarrow \max(0, 1 - N_t/N_{\max}) \cdot A \cdot \mathcal{N}(z)$
      2. Implicit Solve: $\mathbf{M}_{FPE} \cdot \mathbf{f}^* = \mathbf{f}_t + \Delta t \cdot S_{new}$
      3. Flux & Cut: $\Phi_t \leftarrow (\sum_{z<b_t^*} \mathbf{f}^* \Delta z)/\Delta t$; Set $\mathbf{f}^*(z) \leftarrow 0$ for $z < b_t^*$
      4. Recycling: $\mathbf{f}_{t+1} \leftarrow \mathbf{f}^* + (1-\epsilon)\Phi_t g(z; u)\Delta t$; Update N_{t+1}

    Phase 4: Learning. Store $(\mathbf{s}_t, \mathbf{u}_t, r_t, \mathbf{s}_{t+1})$ in $\mathcal{D}$.
      Update Critic θ (Bellman) and Actor ϕ (Max Entropy $J = \mathbb{E}[Q - \alpha \log \pi]$).
  end for
end for
return Optimized Policy π_ϕ^*

Software Stack. The solution algorithm was implemented using **Python 3.10** and **PyTorch 2.0.1** with CUDA 11.8 support. The Mean-Field Game solver and the Soft Actor-Critic (SAC) agent were built upon the *Gymnasium* interface standards and the *Stable Baselines 3* library.

D.4 Hyperparameter Configuration and Reward Shaping

To ensure the stability of the Spectral SAC algorithm and accurate calibration against the empirical moments, we employed specific reward shaping coefficients. The training process utilized the Stable Baselines 3 implementation of Soft Actor-Critic. Table 4 details the key hyperparameters for both the Baseline and Clickbait experiments (10 seeds $\times$ 500,000 steps each).

E Robustness Check: Scale Invariance and Algorithmic Overheating

To rigorously verify that the "Algorithmic Liquidity Trap" identified in Experiment A is not a numerical artifact of lower absolute reward weights, we performed a normalization experiment (Experiment C).

Experimental Design. We constructed a scenario that matches the *Total Weight Budget* of the Baseline (≈ 20) but retains the *Monetization Ratio* of the Ad-Centric regime ($\mathcal{R} = 1$).

- **Baseline:** $\lambda_1 = 20.0, \lambda_2 = 0.01$ (Total ≈ 20).

Table 4: Reward Function Coefficients and Hyperparameters

Parameter	**Baseline**	**Clickbait**	**Description**
Experiment-Specific Parameters (Matching Eq. 21)			
λ_1	20.0	5.0	Weight on retention utility flow
λ_2	0.1	5.0	Weight on swipe flux
Flux floor	0.0	1.5	Minimum flux reward threshold
Flux floor penalty	0.0	1.5	Penalty for falling below flux floor
Flux bonus	0.0	2.0	Bonus for exceeding flux floor
c_{ctrl}	2.0	3.0	Control cost scaling coefficient
c_{smooth}	5.0	6.0	Spectral smoothness penalty
Shared Structural Parameters			
ξ_1	1.0		Retention control cost ($\frac{\xi_1}{2}\|a\|^2$)
ξ_2	0.5		Matching investment cost ($\frac{\xi_2}{2}u^2$)
ψ_0	0.01		Acquisition cost coefficient
SAC Training Configuration (Both Experiments)			
Total Steps	500,000		Per seed
Seeds	10		Independent runs
Learning Rate	3×10^{-4}		Actor & Critic (Adam)
Batch Size	256		Experience replay
γ	0.99		Discount factor

Note: λ_1, λ_2 correspond to the objective function in Eq. (21). The flux floor, penalty, and bonus are reward-shaping terms that incentivize the Clickbait agent to maintain minimum traffic throughput. c_{ctrl} and c_{smooth} are numerical regularization coefficients that prevent degenerate policies; they do not appear in the theoretical model.

- **Experiment A (Original):** $\lambda_1 = 5.0, \lambda_2 = 5.0$ (Total $= 10$).
- **Experiment C (Robustness):** $\lambda_1 = 10.0, \lambda_2 = 10.0$ (Total $= 20$).

Results and Interpretation. The results are visualized in Figure 10.

1. **Validation of the Trap (Flux Consistency):** The equilibrium swipe flux Φ^* in Experiment C converged to a value consistent with Experiment A, representing a multiple-fold increase over the Baseline. This confirms that the high-velocity trap is a structural equilibrium determined by the ratio $\mathcal{R}$, robust to parameter scaling.

2. **Algorithmic Overheating (Population Drop):** The steady-state population N^* in Experiment C was lower than in Experiment A, a structurally consistent outcome of *Relative Cost Reduction*. In the objective function $J = \lambda_1 z + \lambda_2 \Phi - \frac{\xi}{2} a^2$, doubling the rewards ($\lambda$) while holding the control cost coefficient constant ($\xi = 1.0$) effectively makes algorithmic intervention "cheaper" relative to the potential payoff. Consequently, the DRL agent adopted a more aggressive "harvesting" policy—pushing the recycling velocity to its limit to extract immediate flux rewards, even at the cost of higher user burnout and a lower carrying capacity. We term this phenomenon *"Algorithmic Overheating,"* where excessive incentive intensity (without corresponding cost constraints) accelerates ecosystem depletion. This aggressive harvesting drives the implicit dwell time τ down even further below the IAB/MRC viewability threshold, indicating that higher reward scales without cost adjustments exacerbate the race to the bottom.

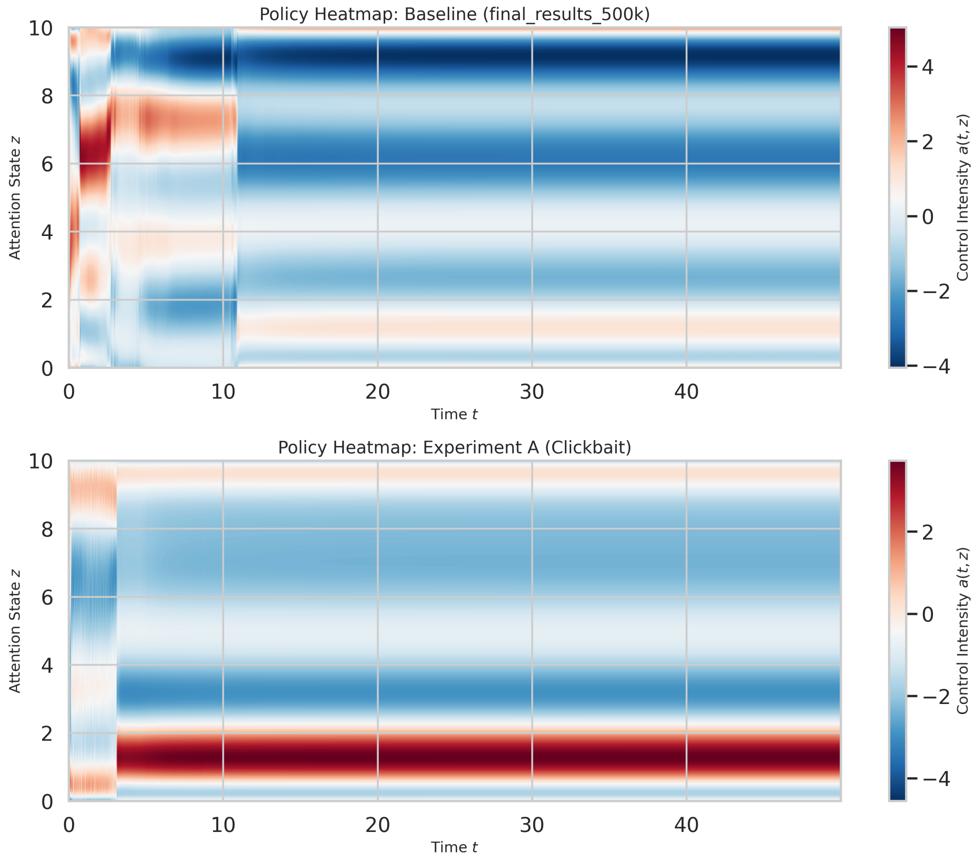


Figure 10: **Robustness Check: Effect of Reward Scale Normalization.** Comparing Experiment A ($\lambda = 5, 5$) with the scaled Experiment C ($\lambda = 10, 10$). **Right Bar:** The Swipe Flux (Φ^*) remains robustly high (> 100), confirming the persistence of the Liquidity Trap regardless of scale. **Left Bar:** The drop in Population (N^*) in Exp C illustrates the "Overheating" effect, where relatively cheaper control costs lead to aggressive over-harvesting.